

The Joint Score of Gaussian and Laplace AR(1) Chains under Ornstein–Uhlenbeck Diffusion

A unified working note:
Bayesian denoising, belief propagation, and inductive bias

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Abstract

This note consolidates, in a single self-contained reference, the working programme on the *joint* score $S(x, t) = \nabla_x \log P_t(x)$ of a Markov chain corrupted by independent Ornstein–Uhlenbeck (OU) diffusion. We deliberately distinguish the joint score from the per-frame marginal score $\partial_{x_u} \log p_t(x_u)$: only the joint object inherits the inter-frame coupling of the Markov prior P_0 , and only the joint object is the right target for the propagator hypothesis at the heart of the thesis programme.

The narrative arc has three acts. Before they begin, two foundational chapters establish the language: the score-as-denoiser identity (Tweedie’s lemma) and the Bayesian view of the chain posterior, which together collapse every score computation to a posterior mean and every posterior mean to a marginalisation problem. The marginalisation is then solved by belief propagation on the chain factor graph: a detailed, from-scratch derivation of the sum–product algorithm, specialised to Gaussian messages, recovers the Kalman filter and the Rauch–Tung–Striebel smoother as the forward and backward passes. **Act I** treats the Gaussian AR(1) benchmark in full closed form: the noised covariance, three equivalent representations of the noised precision, the spectral preservation theorem, the band-fill lifecycle, and the $O(K)$ belief-propagation / Kalman smoother that recovers the exact joint score. **Act II** is the Laplace warm-up at $K = 1$, where Gaussianity is broken while Markovianity is structurally absent; the joint score acquires nonlinearity but Tweedie’s identity carries through unchanged. **Act III** promotes Laplace innovations to $K \geq 2$, retaining the chain factor graph: belief propagation is still exact in principle, the forward message admits a clean closed form, the backward message exhibits a fundamental causal asymmetry, and the remaining structural questions are listed explicitly. The note closes with an information-theoretic argument for why the factor-graph parametrisation should be a more informative prior with respect to a black-box neural approximator: it encodes the Markov prior as architectural inductive bias, in analogy with the role of locality and weight-sharing in convolutional networks. Every quantitative claim carries either an **[Established]** tag (proved and numerically verified at the tolerances of `numerical_audit.py`) or an **[Open]** tag.

Contents

1	Introduction and Motivation	2
2	Setup, notation, and the OU forward channel	4
3	The score is a denoiser: Tweedie’s identity	7
3.1	Three readings of the same identity	9
3.2	The tilted-measure view of the posterior mean	10

3.3	Score matching as denoising regression	12
4	Bayesian view of the chain posterior	13
4.1	Posterior factorisation	14
4.2	The marginalisation problem and where the $O(K^2)$ cost comes from	14
5	Factor graph and tree structure	15
6	Belief propagation on the chain	16
6.1	Sum-product on the chain	16
6.2	Complexity	18
6.3	The Gaussian case: from BP to Kalman	18
7	Act I — Gaussian AR(1) benchmark	18
7.1	Clean covariance and tridiagonal precision	18
7.2	Noised covariance and joint score	19
7.3	Three equivalent representations of Q_t	20
7.4	Lifecycle of the precision matrix	21
7.4.1	The band-fill theorem (corrected)	21
7.4.2	Asymptotic exponential return to identity	23
7.5	The full-chain BP score recovers the matrix score	23
8	Act II — Laplace warm-up at $K = 1$	24
8.1	The joint score acquires nonlinearity	25
8.2	Fourier representation: convolution becomes a product	26
8.3	The curvature is x -dependent	28
9	Act III — Laplace AR(1) at $K \geq 2$	29
9.1	The Markov-structured form of the joint score	29
9.2	The boundary message has a clean closed form	30
9.3	General (non-boundary) messages have no clean form	30
9.4	Innovation coordinates: prior factorises, likelihood densifies	30
9.5	What the structure buys (and what is open)	32
10	Numerical audit summary	32
11	Closing argument: the factor graph as inductive bias	32
11.1	The intuition, made precise	33
11.2	The convolutional analogy	33
11.3	Open programme directions	34
	Reproducibility	34
	References	34

1 Introduction and Motivation

What generative diffusion is. Generative diffusion is a very active field of research. In a nutshell, it aims at generating artificial data $a \in \mathbb{R}^d$ drawn from a distribution $P_0(a)$ which is not known explicitly, but for which we have a database of n examples a^μ , $\mu \in \{1, \dots, n\}$, supposed to be iid samples of P_0 . Notice that P_0 itself is usually not well defined (think of generating images of non-existent celebrities). As we will see, generative diffusion allows one to train a system with a two-step procedure: first *noising* the data through a diffusion process

in $\mathbb{R}^d$, then *denoising* the result through another diffusion process with a guiding force field — the *score* — trained from the data, so that the system generates fresh fake data with properties similar to the originals [1, 8, 11, 13, 18, 19]. This is presently the state of the art in image, text, and video generation. The same formalism can also be used to estimate the likelihood $\log P_0(x)$ of a new point x , using the score obtained from training the generative model [19].

Our setting: dynamic objects. This thesis studies score-based diffusion models applied to *dynamic objects*: data that arise as ordered sequences with a non-trivial internal time, the canonical example being a short video clip. A dynamic object is a sequence of K frames $a_0, a_1, \dots, a_{K-1} \in \mathbb{R}^d$, indexed by internal time $u = 0, \dots, K-1$, drawn from a clean joint distribution $P_0(a_0, \dots, a_{K-1})$ that encodes the temporal dependence of the data. Throughout this thesis we assume that the prior has *Markov structure*: consecutive frames are coupled through a transition kernel $M(a_{u+1} | a_u)$, so that

$$P_0(a_0, \dots, a_{K-1}) = p_0(a_0) \prod_{u=0}^{K-2} M(a_{u+1} | a_u). \quad (1)$$

The diffusion paradigm corrupts the data by applying an independent Ornstein–Uhlenbeck (OU) process to each frame. As we derive in Section 2, the resulting forward channel at noise time t is the affine Gaussian map $x_u = \mu a_u + \sqrt{\Delta_t} \xi_u$ with $\mu = e^{-t}$, $\Delta_t = 1 - e^{-2t}$, and $\xi_u \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Sampling from P_0 reduces, by Anderson’s time-reversal theorem [2, 7], to running a reverse-time SDE whose drift contains the *joint score field*

$$\mathbf{S}(x, t) := \nabla_x \log P_t(x_0, \dots, x_{K-1}) \in \mathbb{R}^{dK}. \quad (2)$$

The k -th block component admits the exact posterior-mean representation

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k | x_0, \dots, x_{K-1}] - x_k}{\Delta_t}, \quad (3)$$

and is emphatically *not* the per-frame marginal score $\nabla_{x_k} \log p_t(x_k)$: the latter integrates out every frame other than x_k , treating the sequence as if it had been generated frame-by-frame independently. The joint score instead couples every frame to the entire trajectory through the full posterior $P(a_0, \dots, a_{K-1} | x_0, \dots, x_{K-1})$.

The central question of this thesis is therefore: *what structure does $\mathbf{S}(x, t)$ inherit from the Markov factorisation (1) of P_0 , and can that structure be exploited to design more principled and computationally efficient score architectures for dynamic data?*

Two observations motivate everything that follows.

Intuition Score and denoising are the same problem

The posterior-mean representation (3) is *Tweedie’s identity*, and it says more than that the score admits a closed form: it says that learning the score $\mathbf{S}(x, t)$ and learning the Bayes-optimal denoiser $x \mapsto \mathbb{E}[a | x]$ are *mathematically the same problem*, related by an affine, prior-independent rescaling. Two consequences follow at once. First, the prior P_0 enters the score *only* through the conditional expectation, so every structural property of $\mathbb{E}[a | x]$ — linearity, locality, factorisation — is inherited by $\mathbf{S}$. Second, the score is linear in x if and only if the Bayes denoiser is linear in x ; the question of when diffusion models admit affine score fields therefore reduces to the classical question of when the posterior mean is affine in the observation, with its clean Gaussian answer.

Intuition Inductive bias is structure, not training data

For a Markov chain prior (1), P_0 factorises over nearest-neighbour pairs and the OU likelihood factorises over single frames (§2). The posterior $P(a \mid x)$ is therefore a tree-structured graphical model, on which belief propagation is exact in $O(K)$ message-passing operations. A neural network parametrising the score blindly re-discovers this graph from data; a factor-graph parametrisation *is* this graph. The inductive bias is the same shift that took convolutional networks from generic MLPs: encoding what we already know about the data into the architecture, instead of spending capacity learning it from samples.

The two ideas are connected: the score is a posterior expectation, the posterior factorises like the prior, and belief propagation is the algorithm that exploits that factorisation. The remainder of this note works the connection out in detail. Section 2 fixes notation and derives the forward channel from the OU SDE; sections 3 and 4 establish the score-as-denoiser identity and the Bayesian view of the chain posterior; section 5 draws the factor graph; section 6 derives belief propagation from sum-product on trees and identifies the Gaussian case with the Kalman smoother; section 7 executes the Gaussian AR(1) benchmark in full closed form; sections 8 and 9 push the Laplace generalisation to $K = 1$ and $K \geq 2$; section 10 reports the audit; section 11 makes the inductive-bias case explicit.

2 Setup, notation, and the OU forward channel

Clean dynamics. We work in the scalar-frame setting $d = 1$; the construction is per-component and extends without change to $d > 1$. A dynamic object is then $a = (a_0, \dots, a_{K-1}) \in \mathbb{R}^K$ drawn from the Markov chain (1). We specialise to two cases:

- *Gaussian AR(1)*: $a_{u+1} = \alpha a_u + \eta_u$ with $\eta_u \sim \mathcal{N}(0, \sigma_\eta^2)$;
- *Laplace AR(1)*: $a_{u+1} = \alpha a_u + \eta_u$ with $\eta_u \sim \text{Laplace}(0, b)$.

In both cases $|\alpha| < 1$ and the law of a_0 is fixed to enforce stationarity. For Gaussian AR(1) this gives the standard relation $\sigma_0^2 = \sigma_\eta^2 / (1 - \alpha^2)$ between the innovation variance and the initial variance, so that $\text{Var}(a_u) = \sigma_0^2$ for every u . When we plot or audit numerically we adopt the unit-variance normalisation $\sigma_0^2 = 1$, so that $\sigma_\eta^2 = 1 - \alpha^2$; the notation table reflects this normalisation.

Forward channel. Each frame is independently corrupted by the variance-preserving OU SDE [15, 21]

$$dx_{u,t} = -x_{u,t} dt + \sqrt{2} dW_{u,t}, \quad x_{u,0} = a_u, \quad u = 0, \dots, K-1, \quad (4)$$

where $\{W_{u,\cdot}\}_{u=0}^{K-1}$ are mutually independent standard Brownian motions. The closed-form transition is obtained by an integrating-factor argument, which we now carry out in full.

Derivation from the OU SDE (4) to the closed-form transition

We drop the frame index u for the moment; the result applies coordinate-wise and the cross-frame independence is reinstated at the end.

STEP 1 (INTEGRATING FACTOR). The deterministic ODE $\dot{x} = -x$ has solution $x(t) = x_0 e^{-t}$. This suggests rescaling the process by e^t to absorb the drift. Define

$$y_t := e^t x_t. \quad (5)$$

We will show that y_t is a pure Itô integral, with no drift; then we read off x_t by inversion.

STEP 2 (APPLY ITÔ'S LEMMA TO $y_t = f(t, x_t)$ WITH $f(t, x) = e^t x$). The partial derivatives are

$$\partial_t f = e^t x, \quad \partial_x f = e^t, \quad \partial_{xx} f = 0. \quad (6)$$

Itô's formula reads $dy_t = \partial_t f dt + \partial_x f dx_t + \frac{1}{2} \partial_{xx} f (dx_t)^2$. Since $\partial_{xx} f = 0$ the Itô correction vanishes, and substituting the SDE (4) for dx_t gives

$$\begin{aligned} dy_t &= e^t x_t dt + e^t (-x_t dt + \sqrt{2} dW_t) \\ &= \underbrace{e^t x_t dt - e^t x_t dt}_{=0} + \sqrt{2} e^t dW_t \\ &= \sqrt{2} e^t dW_t. \end{aligned} \quad (7)$$

The drift cancels by construction — this is the entire purpose of the rescaling. The transformed process y_t is a pure Wiener integral.

STEP 3 (INTEGRATE FROM 0 TO t). The initial condition is $y_0 = e^0 x_0 = a$. Integrating (7),

$$y_t = a + \sqrt{2} \int_0^t e^s dW_s. \quad (8)$$

Multiplying both sides by e^{-t} and using $x_t = e^{-t} y_t$,

$$x_t = e^{-t} a + \sqrt{2} \int_0^t e^{-(t-s)} dW_s. \quad (9)$$

This is the explicit solution: a deterministic exponential decay of the initial condition, plus a stochastic integral against the OU kernel $e^{-(t-s)}$.

STEP 4 (IDENTIFY THE LAW OF THE STOCHASTIC INTEGRAL). Write

$$I_t := \sqrt{2} \int_0^t e^{-(t-s)} dW_s. \quad (10)$$

Since the integrand $s \mapsto \sqrt{2} e^{-(t-s)}$ is deterministic, I_t is a Wiener integral of a deterministic function, hence centred Gaussian. Its variance is given by Itô's isometry, $\mathbb{E} \left[\left(\int_0^t f(s) dW_s \right)^2 \right] = \int_0^t f(s)^2 ds$, so

$$\text{Var}(I_t) = \int_0^t (\sqrt{2} e^{-(t-s)})^2 ds = 2 \int_0^t e^{-2(t-s)} ds. \quad (11)$$

Evaluate the remaining integral by the substitution $u = t - s$, $du = -ds$, which sends $s = 0 \rightarrow u = t$ and $s = t \rightarrow u = 0$:

$$\int_0^t e^{-2(t-s)} ds = \int_t^0 e^{-2u} (-du) = \int_0^t e^{-2u} du = \frac{1 - e^{-2t}}{2}. \quad (12)$$

Multiplying by 2,

$$\text{Var}(I_t) = 1 - e^{-2t}. \quad (13)$$

STEP 5 (STANDARDISE). Define the noise variance

$$\Delta_t := 1 - e^{-2t}, \quad (14)$$

and write $I_t = \sqrt{\Delta_t} \xi$ with $\xi \sim \mathcal{N}(0, 1)$. Combining with (9) gives the closed-form transition:

$$x_{u,t} = \mu a_u + \sqrt{\Delta_t} \xi_u, \quad \mu := e^{-t}, \quad \xi_u \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1). \quad (15)$$

STEP 6 (INDEPENDENCE ACROSS FRAMES). Because the SDEs (4) are driven by independent Brownian motions $\{W_{u,\cdot}\}_{u=0}^{K-1}$, the noises $\{\xi_u\}_{u=0}^{K-1}$ appearing in (15) are mutually independent. The full forward map $a \mapsto x$ is therefore an iid affine Gaussian channel applied to each coordinate.

WHY “VARIANCE-PRESERVING”. For data of unit clean variance $\text{Var}(a) = 1$, the channel (15) yields $\text{Var}(x_t) = \mu^2 \cdot 1 + \Delta_t = e^{-2t} + (1 - e^{-2t}) = 1$ for all $t \geq 0$. The marginal variance of x_t is exactly preserved, which is the standard sense in which (4) is called the variance-preserving SDE (as opposed to “variance-exploding” pure Brownian motion). $\square$

Likelihood factorisation. Since each frame’s OU process is independent of the others, the likelihood factorises across frames:

$$p(x | a) = \prod_{k=0}^{K-1} p_t(x_k | a_k), \quad p_t(x_k | a_k) = \mathcal{N}(x_k; \mu a_k, \Delta_t). \quad (16)$$

This is the second structural feature of the model: the prior is Markov along u , the likelihood is product-form across u . The non-trivial inter-frame coupling of the posterior $P(a | x)$ therefore comes *entirely* from the prior; the channel itself introduces none.

The joint score, formally. Combining (1) and (16), the noised joint density of (2) admits the integral representation

$$P_t(x) = \int_{\mathbb{R}^K} P_0(a) p(x | a) da = \int_{\mathbb{R}^K} P_0(a) \prod_{k=0}^{K-1} \mathcal{N}(x_k; \mu a_k, \Delta_t) da. \quad (17)$$

The k -th component of the joint score (2) is the partial derivative $S_k(x, t) = \partial_{x_k} \log P_t(x)$, taken holding *all other frames* $x_{j \neq k}$ fixed. This is the technical content of the joint-versus-marginal distinction: S_k is a function of the entire noisy trajectory, not of x_k alone.

Bayesian view : the Fokker–Planck view of the forward process

The forward channel (4) can equivalently be characterised by the evolution of the density $P_t(x)$ itself, which solves the *Fokker–Planck equation* [13, §2]

$$\partial_t P_t(x) = \nabla_x \cdot (x P_t(x)) + \Delta_x P_t(x). \quad (18)$$

The first term is the drift contribution from the restoring force $-x$; the second is the diffusion contribution from the white noise. The integral form (17) is the explicit solution of (18) with initial condition $P_{t=0} = P_0$: the Gaussian factor $\mathcal{N}(x_k; \mu a_k, \Delta_t)$ is precisely the Green’s function of the OU Fokker–Planck operator. Two facts will matter later: at $t \rightarrow \infty$, the steady-state solution of (18) is the standard Gaussian $P_\infty(x) = (2\pi)^{-K/2} e^{-\|x\|^2/2}$ (the “easy distribution” from which reverse-time sampling starts); and the $\nabla_x \log P_t$ gradient that appears as a correction to the drift when one time-reverses the Fokker–Planck equation is exactly the score [2, 13, §3.1] which the rest of this note studies.

Notation.

Symbol	Meaning	Note
K	sequence length	
a_k	clean frame at internal time k	
x_k	noisy observation at t	
α	AR(1) coefficient	$ \alpha < 1$
σ_η	Gaussian innovation std	$\sqrt{1 - \alpha^2}$ at $\sigma_0^2 = 1$
σ_0	initial std of a_0	stationary: $\sigma_0^2 = \sigma_\eta^2 / (1 - \alpha^2)$
μ_0	initial mean of a_0	zero in the stationary case
b	Laplace innovation scale	
$\mu = e^{-t}$	OU signal-attenuation factor	
$\Delta_t = 1 - e^{-2t}$	OU noise variance	
P_0, p_0, M	prior, frame-0 marginal, transition	
P_t, p_t	noised joint and marginal	
μ_a	prior mean vector	$\mu_a[k] = \alpha^k \mu_0$
Σ_0	joint clean covariance	Gaussian case
$Q_0 = \Sigma_0^{-1}$	joint clean precision	tridiagonal for AR(1)
$\Sigma_t = \mu^2 \Sigma_0 + \Delta_t I$	joint noised covariance	
$Q_t = \Sigma_t^{-1}$	joint noised precision	
$S(x, t)$	joint score $\nabla_x \log P_t(x)$	
$s(x_u, u, t)$	per-frame marginal score	<i>no inter-correlations</i>

3 The score is a denoiser: Tweedie's identity

The fundamental identity that organises every score computation in this note is Tweedie's lemma, in its joint form [6, 20]. Its content is simple and remarkable: the score field is an affine map of the Bayes-optimal denoiser $\mathbb{E}[a \mid x]$. The prior P_0 enters only through this conditional expectation; the Gaussian channel contributes only the affine constants $\mu = e^{-t}$ and $\Delta_t = 1 - e^{-2t}$. Computing the score and computing the Bayesian denoiser are therefore the same problem; in the OU setting of this thesis the same identity appears in [13, §3] and is the foundation of denoising score matching [22].

Theorem 1 (Joint Tweedie identity, [Established]; cf. [6, 20]). *Let P_0 be any prior on $\mathbb{R}^K$, let $t > 0$, and write the joint noised density induced by the forward channel (15) as*

$$P_t(x) = \int_{\mathbb{R}^K} P_0(a) p(x \mid a) da, \quad p(x \mid a) = \prod_{k=0}^{K-1} \frac{1}{\sqrt{2\pi\Delta_t}} \exp\left(-\frac{(x_k - \mu a_k)^2}{2\Delta_t}\right). \quad (19)$$

Then the joint score $S(x, t)(x, t) := \nabla_x \log P_t(x)$ admits, component-wise, the two equivalent forms

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k \mid x] - x_k}{\Delta_t} = -\frac{\mathbb{E}[\xi_k \mid x]}{\sqrt{\Delta_t}}, \quad k = 0, \dots, K-1, \quad (20)$$

where the conditional expectation is taken under the posterior $P(a \mid x)$, and ξ is the standard Gaussian noise of the channel decomposition $x = \mu a + \sqrt{\Delta_t} \xi$.

Derivation 1: from the channel to Tweedie

The proof structure follows [20, §2], adapted from the pure Brownian channel to the variance-preserving OU channel (15). The score is, by definition, the gradient of a log-density, and the log-gradient of any positive function f satisfies the elementary identity

$$\nabla_x \log f(x) = \frac{\nabla_x f(x)}{f(x)}. \quad (21)$$

Applied to P_t , this turns the computation of the score into the computation of $\partial_{x_k} P_t(x)$ divided by $P_t(x)$, which is what we now do.

Start from the integral expression (19) for $P_t(x)$ and differentiate with respect to x_k . Because the prior $P_0(a)$ does not depend on x , the derivative passes inside the integral and acts only on the Gaussian likelihood $p(x | a)$:

$$\partial_{x_k} P_t(x) = \int_{\mathbb{R}^K} P_0(a) \partial_{x_k} p(x | a) da. \quad (22)$$

The Gaussian likelihood is itself an exponential of a quadratic in x_k , so its x_k -derivative is explicit. Writing $\log p(x | a) = -\frac{(x_k - \mu a_k)^2}{2\Delta_t} + (\text{terms independent of } x_k)$ and differentiating,

$$\partial_{x_k} p(x | a) = -\frac{x_k - \mu a_k}{\Delta_t} p(x | a). \quad (23)$$

The interpretation is already visible at this point: the gradient of the Gaussian channel is the channel itself, multiplied by the “displacement from the channel mean” ($x_k - \mu a_k$), with the expected minus sign and the variance Δ_t in the denominator.

Substitute (23) into (22):

$$\partial_{x_k} P_t(x) = -\frac{1}{\Delta_t} \int_{\mathbb{R}^K} (x_k - \mu a_k) P_0(a) p(x | a) da. \quad (24)$$

The factor x_k does not depend on a and can be pulled outside the integral; the remaining integral of $P_0(a) p(x | a)$ is, by (19), just $P_t(x)$. The integral involving a_k is the object we must understand:

$$\partial_{x_k} P_t(x) = -\frac{1}{\Delta_t} \left[x_k P_t(x) - \mu \int_{\mathbb{R}^K} a_k P_0(a) p(x | a) da \right]. \quad (25)$$

This is the point at which Bayes’ theorem enters. The joint density of the clean–noisy pair is $P_0(a) p(x | a)$, and dividing by its marginal $P_t(x)$ gives the posterior:

$$P(a | x) = \frac{P_0(a) p(x | a)}{P_t(x)}. \quad (26)$$

Dividing $\partial_{x_k} P_t(x)$ by $P_t(x)$ is therefore the same as taking the expectation of $(x_k - \mu a_k)$ under $P(a | x)$, up to the factor $-1/\Delta_t$:

$$\frac{\partial_{x_k} P_t(x)}{P_t(x)} = -\frac{1}{\Delta_t} \int_{\mathbb{R}^K} (x_k - \mu a_k) P(a | x) da = -\frac{x_k - \mu \mathbb{E}[a_k | x]}{\Delta_t}. \quad (27)$$

In the last step we used linearity of the integral, the fact that $P(a | x)$ is a probability measure (so that x_k integrates out to x_k), and the definition of the conditional expectation $\mathbb{E}[a_k | x] = \int a_k P(a | x) da$.

By the log-gradient identity (21), the left-hand side of (27) is exactly the k -th component $S_k(x, t)$ of the joint score. Rearranging,

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k | x] - x_k}{\Delta_t}, \quad (28)$$

which is the first form announced in (20).

The second, “noise-form” identity is now a one-line rewriting of the first. The forward channel decomposes as $x = \mu a + \sqrt{\Delta_t} \xi$, with $\xi \sim \mathcal{N}(0, I)$ independent of a ; equivalently,

on the joint space, $\xi_k = (x_k - \mu a_k)/\sqrt{\Delta_t}$. Conditioning on x and taking expectations of both sides — noting that x_k is a function of x and so passes through the conditional expectation unchanged — yields

$$\mathbb{E}[\xi_k | x] = \frac{x_k - \mu \mathbb{E}[a_k | x]}{\sqrt{\Delta_t}}. \quad (29)$$

Comparing with (28), the right-hand side equals $-\sqrt{\Delta_t} S_k(x, t)$, so

$$S_k(x, t) = -\frac{\mathbb{E}[\xi_k | x]}{\sqrt{\Delta_t}}, \quad (30)$$

which is the second form in (20). $\square$

3.1 Three readings of the same identity

The two equivalent forms of (20) carry three operational readings. The first reading is the *denoiser identity*: the score is, up to known channel constants, the affine correction that converts the noisy observation into the Bayes-optimal estimate of the clean signal. The second reading is the *posterior expected noise* interpretation, which is what the reverse-time SDE actually adds back during sampling. The third reading — the one that organises the rest of this note — is that the prior P_0 enters the score only through the conditional expectation $\mathbb{E}[a | x]$, so every structural property of the score (linearity, factorisation, sparsity) is inherited from the structure of the Bayesian denoiser.

Bayesian view : the score *is* the optimal denoiser

Solving (20) for the conditional expectation,

$$\mathbb{E}[a | x] = \frac{x + \Delta_t S(x, t)}{\mu}. \quad (31)$$

The Bayes-optimal denoiser at every time t is an affine function of the score with known coefficients. Knowing the score and knowing the optimal denoiser are exactly the same problem; everything below is ultimately a strategy for computing $\mathbb{E}[a | x]$ and reading off $S(x, t)$ from (31).

Bayesian view : the score is the posterior expected noise

The second form in (20) reads $S(x, t) = -\mathbb{E}[\xi | x]/\sqrt{\Delta_t}$. Up to the channel scaling $\sqrt{\Delta_t}$, the score is the conditional expectation of the corruption noise given the observation. The reverse-time generative SDE (35), derived below, therefore literally “adds back twice the expected noise”: diffusion sampling is the sequential cancellation of the corruption that the forward channel injected [2, 8, 19].

Derivation from forward FPE to backward SDE: where the score enters as drift

The reverse-time SDE alluded to above is a special case of Anderson’s general theorem on time reversal of diffusions [2, 7]. We sketch the OU case to make explicit how the score appears [13, §3.1].

Step 1 (write the FPE in reversed time). The forward Fokker–Planck (18) reads $\partial_t P_t = \nabla \cdot (x P_t) + \Delta P_t$ and governs the evolution of the noised density forward in time. Define the reverse-time variable $\tau := t_f - t$ and the reverse-time density $\tilde{P}_\tau(x) := P_{t_f - \tau}(x)$. Then $\partial_\tau \tilde{P}_\tau = -\partial_t P_{t_f - \tau}$, which after substituting (18) gives

$$\partial_\tau \tilde{P}_\tau(x) = -\nabla \cdot (x \tilde{P}_\tau(x)) - \Delta \tilde{P}_\tau(x). \quad (32)$$

This is *not* the FPE of a forward Langevin equation: the diffusion term has the wrong sign.

Step 2 (replace one Laplacian by a score-drift). Note that $\nabla \tilde{P}_\tau = (\nabla \log \tilde{P}_\tau) \tilde{P}_\tau$, so $\Delta \tilde{P}_\tau = \nabla \cdot ((\nabla \log \tilde{P}_\tau) \tilde{P}_\tau)$. Adding and subtracting one copy of $\Delta \tilde{P}_\tau$ – i.e. using the identity $A = 2A - A$ with $A = \Delta \tilde{P}_\tau$ and substituting the score expression for the first copy – rewrites the wrong-sign Laplacian as

$$-\Delta \tilde{P}_\tau = \underbrace{-2 \nabla \cdot ((\nabla \log \tilde{P}_\tau) \tilde{P}_\tau)}_{\text{a new drift term}} + \underbrace{\Delta \tilde{P}_\tau}_{\text{Laplacian, now with the right sign}}. \quad (33)$$

No new information has entered (33): it is the identity $A = 2A - A$ with one of the two A ’s rewritten via $\Delta P = \nabla \cdot ((\nabla \log P) P)$. Its purpose is *purely structural* – it splits the wrong-sign diffusion of Step 1 into a drift term plus a right-sign diffusion term, exposing exactly which extra drift is needed to convert the reversed FPE into that of a forward Langevin equation.

Step 3 (collect). Substituting (33) into (32) gives the FPE

$$\partial_\tau \tilde{P}_\tau(x) = -\nabla \cdot [(x + 2 \nabla \log \tilde{P}_\tau(x)) \tilde{P}_\tau(x)] + \Delta \tilde{P}_\tau(x). \quad (34)$$

The right-hand side is now the FPE of a forward Langevin equation in the variable τ , with drift $x + 2 \nabla \log \tilde{P}_\tau(x)$ and the standard diffusion. Reading off the corresponding SDE,

$$dx = (x + 2S(x, t_f - \tau)) d\tau + \sqrt{2} d\tilde{W}_\tau, \quad (35)$$

where $\tilde{W}$ is a standard Brownian motion driving the reverse-time process and $S(x, t) := \nabla_x \log P_t(x)$ is the score. The drift now *adds* x (rather than $-x$, as in the forward SDE (4)) precisely because we are running in reversed time $\tau = t_f - t$; the $+2S$ correction is the piece of (33) that splits off as a drift. Equation (35) is the reverse-time generative SDE, in the reversed-time variable τ .

Intuition : nonlinearity tracks the prior

For Gaussian P_0 , the conditional expectation $\mathbb{E}[a | x]$ is affine in x (Gaussian conditioning), so the score is affine. For non-Gaussian P_0 , the conditional expectation is a generally nonlinear function of x , and so is the score. This is the only mechanism by which the prior enters the score, and the only mechanism by which the score acquires nonlinearity. Section 8 verifies this on the simplest non-Gaussian case ($K = 1$, Laplace prior).

3.2 The tilted-measure view of the posterior mean

A second perspective on Tweedie’s identity, due to Mézard [13, § 3.1.1], makes very explicit how the observation x shapes the posterior over the clean signal. We work in the single-frame case for clarity; the argument extends component-wise.

Theorem 2 (Tilted-measure representation of the posterior mean, [Established]). *For the OU forward channel and any prior p_0 ,*

$$\mathbb{E}[a \mid x] = \frac{1}{Z_{x,t}} \int_{\mathbb{R}} a p_0(a) \exp\left(-\frac{\mu^2}{2\Delta_t} (a - \tilde{x})^2\right) da, \quad \tilde{x} := \frac{x}{\mu} = e^t x, \quad (36)$$

where $Z_{x,t}$ is the normalising integral.

Derivation 2: rewriting Bayes in terms of a quadratic pinning

By Bayes' theorem, $P(a \mid x) \propto p_0(a) p(x \mid a)$ with $p(x \mid a) \propto \exp(-(x - \mu a)^2 / (2\Delta_t))$. Inside the exponent, complete the square in a :

$$-\frac{(x - \mu a)^2}{2\Delta_t} = -\frac{\mu^2}{2\Delta_t} a^2 + \frac{\mu x}{\Delta_t} a + \text{const}_a = -\frac{\mu^2}{2\Delta_t} (a - \tilde{x})^2 + \text{const}_a, \quad (37)$$

with $\tilde{x} = x/\mu = e^t x$. The constants drop into the normalisation $Z_{x,t}$; the remaining a -dependent factor is a Gaussian pinning centred at $\tilde{x}$ with strength $\mu^2/\Delta_t = e^{-2t}/\Delta_t$. Multiplying by $p_0(a)$ and normalising gives (36).

Intuition : $\tilde{x} = e^t x$ is the projected clean trajectory; the pinning strength tells us how to trust it

(36) says $\mathbb{E}[a \mid x]$ is the average of a under a measure obtained from the prior p_0 by *tilting* it with a quadratic pinning centred at $\tilde{x}$. Two limiting behaviours are obvious from this form [13, § 3.1.1]:

- **Small t : pinning dominates.** The pinning strength μ^2/Δ_t diverges like $1/(2t)$. The tilted measure concentrates at $\tilde{x}$, so $\mathbb{E}[a \mid x] \rightarrow \tilde{x} = x$. The observation x is trustworthy and the score $S(x, t) \rightarrow 0$: no denoising is needed.
- **Large t : pinning vanishes.** The strength $\mu^2/\Delta_t \rightarrow 0$, the tilt becomes flat, and $\mathbb{E}[a \mid x] \rightarrow \int a p_0(a) da$ (the prior mean – here zero). The observation carries essentially no information; the optimal estimate falls back on the prior.

The whole nontrivial regime of the diffusion model lives at intermediate t , where the pinning is comparable to the characteristic spread of p_0 and the tilted measure is genuinely deformed by both. This is also the regime in which the *structure* of p_0 becomes visible in the score: the tilt amplifies any region where p_0 places mass near $\tilde{x}$, and suppresses the rest.

Bayesian view : when P_0 has a smooth potential, the score is an expectation of the original force

If $p_0(a) \propto e^{-V_0(a)}$ for a smooth potential V_0 , the score admits the equivalent expression [13, § 3.1.1]

$$S(x, t) = e^t \langle F_0(a) \rangle_{\text{tilt}}, \quad F_0(a) := -\nabla V_0(a), \quad (38)$$

where the expectation is taken under the tilted measure (36) and F_0 is the *clean-time force* that would drive a Langevin sampler of p_0 .

Derivation 38: rewriting the posterior-mean residual as a tilted expectation of $\nabla \log p_0$

Tweedie (20) gives $S(x, t) = (\mu \mathbb{E}[a \mid x] - x)/\Delta_t$. Differentiating the log of (36) – the tilted-measure representation – in the direction of a and integrating against p_0 the partial derivative ∂_a gives, by integration by parts, an equivalent form. Concretely:

Step 1. Write $p_0(a) = Z_0^{-1} e^{-V_0(a)}$ and recall that the tilted measure $p_{\text{tilt}}(a) \propto p_0(a) e^{-\mu^2(a-\tilde{x})^2/(2\Delta_t)}$ satisfies

$$\partial_a \log p_{\text{tilt}}(a) = \partial_a \log p_0(a) - \frac{\mu^2}{\Delta_t} (a - \tilde{x}) = -\nabla V_0(a) - \frac{\mu^2}{\Delta_t} (a - \tilde{x}). \quad (39)$$

Step 2. A normalised density satisfies $\mathbb{E}_{\text{tilt}}[\partial_a \log p_{\text{tilt}}(a)] = 0$ (integration by parts on a function decaying at infinity). Taking the expectation of both sides of Step 1 under p_{tilt} ,

$$0 = \langle F_0(a) \rangle_{\text{tilt}} - \frac{\mu^2}{\Delta_t} (\mathbb{E}_{\text{tilt}}[a] - \tilde{x}) = \langle F_0(a) \rangle_{\text{tilt}} - \frac{\mu^2}{\Delta_t} (\mathbb{E}[a | x] - x/\mu), \quad (40)$$

using $\tilde{x} = x/\mu$ and the identification of the tilted expectation with the posterior expectation $\mathbb{E}[a | x]$ (the tilt is exactly the channel likelihood times the prior).

Step 3. Solve for the posterior-mean residual:

$$\frac{\mu \mathbb{E}[a | x] - x}{\Delta_t} = \frac{1}{\mu} \langle F_0(a) \rangle_{\text{tilt}} = e^t \langle F_0(a) \rangle_{\text{tilt}}, \quad (41)$$

since $1/\mu = e^t$. The left-hand side is exactly the score by Tweedie (20), which gives (38). $\square$

Intuition : the score is, up to e^t , the clean-time force averaged where the channel believes a lives

(38) is a clean physical reading. At any noise level t , the score evaluates the original Langevin force $F_0 = -\nabla V_0$ but *not at the bare prior measure*: instead at the tilted measure, which pinches p_0 toward the deconvolved observation $\tilde{x} = e^t x$. The expansion factor $e^t = 1/\mu$ compensates for the channel-induced contraction of a onto x : had the channel not attenuated the signal, no rescaling would be needed.

3.3 Score matching as denoising regression

The denoiser reading of Tweedie's identity has an operational counterpart at the level of *training*: the score-matching loss [9] is, up to an additive constant, the mean-squared error of a noise predictor. This is the foundational result of Vincent [22]; we give the explicit derivation in Mézard's form [13, §3.1.1].

Theorem 3 (Score matching equals denoising MSE, [Established]). *Let $\hat{S}(x, t)$ be any candidate score field. The exact score-matching loss*

$$L_{\text{exact}}(\hat{S}) = \int_0^{t_f} dt \mathbb{E}_{x \sim P_t} \left\| \hat{S}(x, t) - \nabla_x \log P_t(x) \right\|^2 \quad (42)$$

is equal, up to a $\hat{S}$ -independent additive constant C , to the denoising loss

$$L_{\text{exact}}(\hat{S}) = \int_0^{t_f} dt \mathbb{E}_{a \sim P_0} \mathbb{E}_{z \sim \mathcal{N}(0, I)} \left\| \hat{S}(\mu a + \sqrt{\Delta_t} z, t) + \frac{z}{\sqrt{\Delta_t}} \right\|^2 + C. \quad (43)$$

Derivation 3: expand the square, identify the cross term, complete it

Step 1 — expand the squared norm. Writing $\nabla_x \log P_t(x) =: S^*(x, t)$, we have

$$L_{\text{exact}} = \int_0^{t_f} dt \mathbb{E}_{x \sim P_t} [\|\hat{S}\|^2 - 2\hat{S} \cdot S^* + \|S^*\|^2]. \quad (44)$$

The third term is independent of $\hat{S}$ and joins the constant C .

Step 2 — the cross term unfolds as an integral against $\nabla_x P_t$. Using $S^* P_t = \nabla_x P_t$,

$$\mathbb{E}_{x \sim P_t} [\hat{S} \cdot S^*] = \int \hat{S}(x, t) S^*(x, t) P_t(x) dx = \int \hat{S}(x, t) \nabla_x P_t(x) dx. \quad (45)$$

Step 3 — substitute the closed form of $\nabla_x P_t$. We computed $\nabla_x P_t$ explicitly in (24): $\nabla_x P_t(x) = -\int (x - \mu a) P_0(a) p(x | a) da / \Delta_t$. Inserting this into (45) and switching the order of integration,

$$\mathbb{E}_{x \sim P_t} [\hat{S} \cdot S^*] = -\mathbb{E}_{a \sim P_0} \mathbb{E}_{x \sim p(\cdot | a)} [\hat{S}(x, t) \cdot (x - \mu a) / \Delta_t]. \quad (46)$$

Step 4 — reparametrise $x = \mu a + \sqrt{\Delta_t} z$. Under the channel, $x = \mu a + \sqrt{\Delta_t} z$ with $z \sim \mathcal{N}(0, I)$. Then $(x - \mu a) / \Delta_t = z / \sqrt{\Delta_t}$, so the cross term becomes

$$\mathbb{E}_{x \sim P_t} [\hat{S} \cdot S^*] = -\mathbb{E}_{a \sim P_0} \mathbb{E}_{z \sim \mathcal{N}(0, I)} [\hat{S}(\mu a + \sqrt{\Delta_t} z, t) \cdot z / \sqrt{\Delta_t}]. \quad (47)$$

Step 5 — recombine. Pulling the same change of variables through the $\|\hat{S}\|^2$ term (which is already an expectation under P_t , realisable as $a \sim P_0$ followed by the channel),

$$L_{\text{exact}} = \int_0^{t_f} dt \mathbb{E}_{a \sim P_0} \mathbb{E}_{z \sim \mathcal{N}(0, I)} \left[\|\hat{S}\|^2 + 2\hat{S} \cdot \frac{z}{\sqrt{\Delta_t}} \right] + C', \quad (48)$$

where C' absorbs the constants from Step 1. Completing the square in $\hat{S}$ and $z / \sqrt{\Delta_t}$ gives (43) with $C = C' - \int_0^{t_f} dt \mathbb{E}_z \|z / \sqrt{\Delta_t}\|^2 = C' - \int_0^{t_f} K / \Delta_t dt$, a fixed constant. $\square$

Intuition : training a score model $\equiv$ training a noise predictor

(43) is the practical face of Tweedie. The loss that defines an optimally-trained score $\hat{S}$ is, term by term, the mean squared error of $-\hat{S}(\mu a + \sqrt{\Delta_t} z, t)$ as a predictor of the standardised noise $z / \sqrt{\Delta_t}$. The standard score-matching pipeline — sample (a, z) pairs, compute $x = \mu a + \sqrt{\Delta_t} z$, regress $-\hat{S}(x, t)$ on $z / \sqrt{\Delta_t}$ — is literally denoising regression in disguise. This is why the score-as-denoiser identity is operational rather than merely suggestive: the two objects are not only equal at the optimum, they are also equal as *loss landscapes*.

4 Bayesian view of the chain posterior

Theorem 1 reduces the score to the Bayesian denoiser. For a chain prior the structure of $P(a | x)$ is the only thing that determines whether that denoiser is computable.

4.1 Posterior factorisation

Proposition 4 (Posterior factorisation, [\[Established\]](#)). *For the chain prior (1) and the iid likelihood (16), the posterior factorises as*

$$P(a \mid x) = \frac{P_0(a)p(x \mid a)}{P_t(x)} \propto p_0(a_0) \prod_{u=0}^{K-2} M(a_{u+1} \mid a_u) \prod_{k=0}^{K-1} p_t(x_k \mid a_k). \quad (49)$$

Derivation : combining prior and likelihood factorisations

Bayes' theorem gives $P(a \mid x) = P_0(a)p(x \mid a)/P_t(x)$. The prior factorisation (1) contributes $p_0(a_0)$ and the $K-1$ transition factors; the likelihood factorisation (16) contributes K local-evidence factors. The denominator $P_t(x)$ does not depend on a and contributes only normalisation.

The right-hand side of (49) has a precise graphical interpretation that we will exploit in section 5 and section 6: each factor involves at most two adjacent variables, so the dependence graph is a chain.

4.2 The marginalisation problem and where the $O(K^2)$ cost comes from

To read off the score component S_k we need the posterior mean $\mathbb{E}[a_k \mid x]$, which only depends on the marginal posterior $p(a_k \mid x)$. Writing da_{-k} for “integrate over every coordinate except a_k ” and applying Proposition 4,

$$p(a_k \mid x) \propto \int \cdots \int \left[p_0(a_0) \prod_{u=0}^{K-2} M(a_{u+1} \mid a_u) \prod_{j=0}^{K-1} p_t(x_j \mid a_j) \right] da_{-k}. \quad (50)$$

Computation : brute-force counting and where it lands at $O(K^2)$

Step 1 — the integrand factorises into nearest-neighbour pieces. Inside the bracket of (50), every factor involves at most two adjacent variables: the prior $p_0(a_0)$ involves only a_0 ; each transition $M(a_{u+1} \mid a_u)$ involves only a_u and a_{u+1} ; each local evidence $p_t(x_j \mid a_j)$ involves only a_j (with x_j constant).

Step 2 — a single a_j -integral is one-dimensional. Fix some $j \neq k$ and integrate a_j first. Only the three factors that involve a_j — two transition factors $M(a_j \mid a_{j-1}), M(a_{j+1} \mid a_j)$ and one local evidence $p_t(x_j \mid a_j)$ — contribute to the integral. Everything else can be pulled outside. So the integration over a_j is a one-dimensional integral, regardless of how large K is.

Step 3 — one marginal requires $(K-1)$ such integrals. To produce $p(a_k \mid x)$ we have to perform $K-1$ such one-dimensional integrations, one for each $j \neq k$.

Step 4 — all K marginals. Repeating Step 3 for every $k = 0, \dots, K-1$ and ignoring the possibility of sharing intermediate work, the total number of one-dimensional integrals is

$$\text{cost}_{\text{BF}} = K \cdot (K-1) = O(K^2). \quad (51)$$

Intuition : where belief propagation finds the missing K factor

The $O(K^2)$ count above is honest *per marginal* ($K-1$ one-dimensional integrals), but is wasteful at the *across-marginal* level: the integral over a_j appears in the computation

Factor graph of the AR(1) chain with OU corruption: a tree (path), so belief propagation is exact

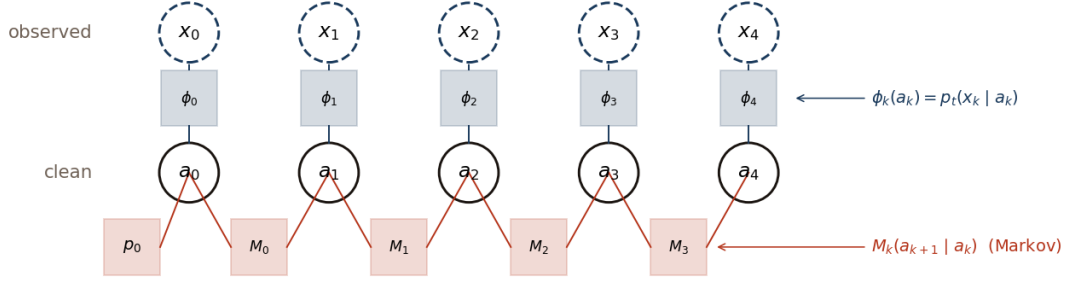


Figure 1: Factor graph of the chain prior and OU likelihood. Solid edges connect clean variables a_k (white circles) to the prior factor p_0 and the Markov transition factors $M_k = M(a_{k+1} | a_k)$ (red squares). Dashed edges connect each clean variable to its local evidence factor $\phi_k(a_k) = p_t(x_k | a_k)$ (navy squares) and through ϕ_k to the observed x_k (dashed circle). The dependence among the a_k is a path; the local evidence factors have degree 1 from the a -side and add no edges among the a_k ; the graph is therefore a tree.

of every marginal $p(a_k | x)$ with $k \neq j$, and the way Step 3 was counted, that integral is repeated $K - 1$ times. Belief propagation keeps every one-dimensional integral as a reusable *message* – a function of one variable, summarising the contribution of one entire half of the chain – and reuses it for every marginal. The total number of integrals collapses from $K(K - 1) = O(K^2)$ to $2(K - 1) = O(K)$: $K - 1$ integrals forward, $K - 1$ integrals backward, then a constant-cost combination per frame. Section 6 makes this precise.

5 Factor graph and tree structure

A *factor graph* [12, 14] is a bipartite graph in which one set of nodes represents random variables and the other set represents factors. Variable node a_k is connected to factor node ϕ if and only if ϕ depends on a_k . For our posterior (49):

- Variable nodes: $a_0, a_1, \dots, a_{K-1}$ (clean), and optionally the observed $x_0, \dots, x_{K-1}$.
- Prior factor: p_0 attached to a_0 (degree 1).
- Transition factors: $M_k \equiv M(a_{k+1} | a_k)$ attached to a_k and a_{k+1} (degree 2).
- Local evidence factors: $\phi_k(a_k) := p_t(x_k | a_k)$ attached to a single variable node a_k (degree 1 from the a -side, with the observed x_k treated as a constant).

The factor graph is drawn in Figure 1.

Proposition 5 (Chain factor graph is a tree, [Established]). *The factor graph of the posterior (49) is a tree.*

Derivation : counting edges and excluding cycles

A tree on V nodes has exactly $V - 1$ edges and no cycles. Our factor graph has $V = K$ variable nodes plus K local-evidence factors plus $K - 1$ transition factors plus 1 prior factor,

hence $V = 3K$. Edges: each transition factor contributes 2 edges ($K - 1$ factors $\Rightarrow 2K - 2$ edges), each local-evidence factor contributes 1 edge from a_k -side ($\Rightarrow K$ edges), the prior contributes 1 edge. Total: $(2K - 2) + K + 1 = 3K - 1 = V - 1$. For acyclicity, observe that the only structure that could form a cycle is among the variable nodes mediated by transition factors; but successive transitions form a path, not a cycle, because each M_k is connected only to its two endpoints and the local-evidence and prior factors are leaves. The graph is therefore connected (each a_k is reachable from a_0 by following transitions) with $V - 1$ edges; it is a tree.

Proposition 6 (Global Markov property, [\[Established\]](#)). *Conditioning on any variable node a_k disconnects the graph into two independent subtrees: the left sub-chain $(a_0, \dots, a_{k-1})$ together with their local evidence $x_0, \dots, x_{k-1}$, and the right sub-chain $(a_{k+1}, \dots, a_{K-1})$ together with their local evidence $x_{k+1}, \dots, x_{K-1}$. The two halves are conditionally independent given a_k .*

Derivation : cut-set argument

Any path between an a_j with $j < k$ and an a_ℓ with $\ell > k$ in the factor graph must pass through a_k (the chain is a path graph; there is no other route). Conditioning on a_k removes a_k as an active node, severing every such path. The two halves share no remaining factor; hence by the global Markov property of factor graphs, they are conditionally independent given a_k .

Intuition : why the tree structure is what makes BP exact

On a tree, marginalisation can be performed by sequential elimination of leaves — absorb a leaf into its unique neighbour, peel it off, repeat. Each elimination step is a one-dimensional integral. The sum-product algorithm [\[12, 16\]](#) formalises this peeling into two passes (forward, backward), each visiting every edge once. On a graph with cycles the same recursion is not exact: information sent into the graph eventually returns, doubly counted. The cycle-free structure of the chain factor graph is exactly the absence of this double-counting — which is why BP is exact here.

6 Belief propagation on the chain

We now derive belief propagation [\[12, 16\]](#) on the chain factor graph in detail and, in the Gaussian case, briefly note its identification with the Kalman filter [\[10\]](#) and the Rauch–Tung–Striebel smoother [\[17\]](#). The whole construction is rebuilt from scratch — earlier internal notes contained a double-counting trap that we explicitly correct.

6.1 Sum-product on the chain

We adopt *Convention A*: the forward message at node k carries the evidence $x_0, \dots, x_{k-1}$ only; the backward message carries $x_{k+1}, \dots, x_{K-1}$ only; the local evidence $p_t(x_k | a_k)$ is inserted at combination time, not inside either recursion. This avoids double-counting any x_k and is the only convention under which the simple combination rule [\(56\)](#) below holds.

Proposition 7 (Convention A messages, [\[Established\]](#)). *Define*

$$\mu_{\rightarrow 0}(a_0) := p_0(a_0), \quad (52)$$

$$\mu_{\rightarrow k+1}(a_{k+1}) := \int M(a_{k+1} \mid a_k) p_t(x_k \mid a_k) \mu_{\rightarrow k}(a_k) da_k, \quad (53)$$

$$\mu_{\leftarrow K-1}(a_{K-1}) := 1, \quad (54)$$

$$\mu_{\leftarrow k-1}(a_{k-1}) := \int M(a_k \mid a_{k-1}) p_t(x_k \mid a_k) \mu_{\leftarrow k}(a_k) da_k. \quad (55)$$

Then for every k ,

$$p(a_k \mid x) \propto \mu_{\rightarrow k}(a_k) p_t(x_k \mid a_k) \mu_{\leftarrow k}(a_k). \quad (56)$$

Derivation 7: peeling the chain at a_k

Start from the joint $P(a, x) = p_0(a_0) \prod_u M(a_{u+1} \mid a_u) \prod_j p_t(x_j \mid a_j)$, and integrate out every a_j with $j \neq k$. By the global Markov property (Proposition 6), this integral splits into two independent halves at a_k : the left half involves $(a_0, \dots, a_{k-1})$ and the evidence $(x_0, \dots, x_{k-1})$ with the prior p_0 ; the right half involves $(a_{k+1}, \dots, a_{K-1})$ and $(x_{k+1}, \dots, x_{K-1})$. The local evidence factor at node k , $p_t(x_k \mid a_k)$, depends only on a_k and on the *constant* x_k ; it factors out of both integrals.

Left half. Define recursively $\mu_{\rightarrow k}(a_k) := \int p_0(a_0) \prod_{u=0}^{k-1} M(a_{u+1} \mid a_u) \prod_{j=0}^{k-1} p_t(x_j \mid a_j) da_0 \cdots da_{k-1}$. Peeling off the leftmost unintegrated variable, a_{k-1} ,

$$\begin{aligned} \mu_{\rightarrow k}(a_k) &= \int M(a_k \mid a_{k-1}) p_t(x_{k-1} \mid a_{k-1}) \\ &\quad \times \left[\int p_0(a_0) \prod_{u=0}^{k-2} M(a_{u+1} \mid a_u) \prod_{j=0}^{k-2} p_t(x_j \mid a_j) da_0 \cdots da_{k-2} \right] da_{k-1} \\ &= \int M(a_k \mid a_{k-1}) p_t(x_{k-1} \mid a_{k-1}) \mu_{\rightarrow k-1}(a_{k-1}) da_{k-1}, \end{aligned} \quad (57)$$

which is exactly the recursion (53). The base case is (52) since when $k = 0$ there is nothing to integrate beyond the prior.

Right half. The analogous argument peeling from the right gives (55)–(54). The base case $\mu_{\leftarrow K-1} \equiv 1$ is uninformative: there is no evidence to the right of the last frame.

Combination. The peeling left, multiplying by the unintegrated local evidence $p_t(x_k \mid a_k)$, and the peeling right give exactly (56), with the proportionality constant being $P_t(x)$.

Remark 1 (The double-counting trap). A natural-looking but *incorrect* alternative (*Convention B*) absorbs the local evidence into the forward message, $\mu_{\rightarrow k}^B(a_k) := \mu_{\rightarrow k}^A(a_k) p_t(x_k \mid a_k)$. Under Convention B the simple combination (56) double-counts x_k (once inside $\mu_{\rightarrow k}^B$ and once at combination time) and must be corrected by dividing by one local evidence factor. Convention A is the one that makes the textbook combination rule correct without correction. The numerical audit (section 10) was decisive in pinning this down: under Convention A the BP score matches the matrix posterior to 10^{-14} ; under naive Convention B the disagreement was order-one.

6.2 Complexity

Computation : BP is $O(K)$, brute force was $O(K^2)$

Each pass ((53) or (55)) computes $K-1$ messages, one per edge, via a single one-dimensional integral each. The combination (56) at every node is a product of three one-variable functions followed by one one-dimensional normalisation – constant cost per node. The total number of one-dimensional integrals is therefore

$$\text{cost}_{\text{BP}} = \underbrace{(K-1)}_{\text{forward}} + \underbrace{(K-1)}_{\text{backward}} + \underbrace{K}_{\text{combine}} = O(K). \quad (58)$$

This is the precise drop from the $O(K^2)$ brute-force counting of (51): BP saves a factor K by storing every intermediate integral as a message and reusing it across all marginals, instead of repeating it once per output marginal.

6.3 The Gaussian case: from BP to Kalman

When P_0 is Gaussian, every factor in the posterior (49) is a Gaussian function of its variables, and the BP recursions (53)–(55) preserve Gaussianity: every message is itself Gaussian, parametrised by a single (m, v) pair per node. The forward recursion then reduces to the predict–update cycle of the *Kalman filter* [10], and the combination of forward and backward passes reduces to the *Rauch–Tung–Striebel smoother* [17]; we do not redo these classical derivations here and refer to standard references [4, 12] for the algebraic details. The audit (§10) verifies our Convention-A implementation `code/bp_score.py` against an independent matrix-form computation of $\Sigma_t^{-1}(x - \mu \mu_a)$ at machine precision over 1600 configurations: in the Gaussian setting the BP score coincides with the closed-form affine score of Theorem 11, exactly as the equivalence “sum–product on a Gaussian chain $\equiv$ Kalman filter and smoother” predicts.

For the non-Gaussian case (§8, §9) Gaussian closure breaks down and the messages must be tracked as one-dimensional functions. The structural content of the recursions (53)–(55) survives unchanged — only the parametrisation does not.

7 Act I — Gaussian AR(1) benchmark

The Gaussian AR(1) chain is the only model in which the joint score admits a complete closed-form treatment. We use it to disentangle two structural properties that the literature often conflates: **Gaussianity**, responsible for the *linearity* of the score, and **Markovianity**, responsible for the *tridiagonal sparsity* of the clean-time precision Q_0 .

7.1 Clean covariance and tridiagonal precision

Proposition 8 (Stationary AR(1) covariance, [Established]; cf. [5]). *For the stationary Gaussian AR(1) chain on K frames,*

$$\Sigma_0[i, j] = \frac{\sigma_\eta^2}{1 - \alpha^2} \alpha^{|i-j|}. \quad (59)$$

Derivation 8: unrolling the recursion

By induction $\mathbb{E}[a_u] = \alpha \mathbb{E}[a_{u-1}] = 0$ for all u , since $\mathbb{E}[\eta_u] = 0$ and a_0 is centred. For $u \geq u'$, the AR(1) recursion unrolls as

$$a_u = \alpha^{u-u'} a_{u'} + \sum_{j=0}^{u-u'-1} \alpha^j \eta_{u-j}. \quad (60)$$

The innovation noise terms η_{u-j} are independent of $a_{u'}$ (they are indexed strictly after u'), so $\mathbb{E}[a_u a_{u'}] = \alpha^{u-u'} \mathbb{E}[a_{u'}^2]$. Stationarity forces $\mathbb{E}[a_{u'}^2] = \sigma_\eta^2 / (1 - \alpha^2)$ for every u' , hence the stated formula. By symmetry the same holds for $u < u'$ with $|u - u'|$ replacing $u - u'$.

Proposition 9 (Tridiagonal clean precision, [Established](#)). *The inverse $Q_0 = \Sigma_0^{-1}$ is exactly tridiagonal:*

$$Q_0[k, k] = \begin{cases} 1/\sigma_\eta^2, & k \in \{0, K-1\}, \\ (1 + \alpha^2)/\sigma_\eta^2, & 0 < k < K-1, \end{cases} \quad Q_0[k, k \pm 1] = -\alpha/\sigma_\eta^2, \quad (61)$$

all other entries identically zero.

Derivation 9: from the joint log-density

The Markov factorisation gives the joint log-density

$$\log P_0(a) = -\frac{a_0^2}{2\sigma_0^2} - \sum_{u=0}^{K-2} \frac{(a_{u+1} - \alpha a_u)^2}{2\sigma_\eta^2} + \text{const.} \quad (62)$$

The matrix Q_0 is the negative Hessian of $\log P_0$ with respect to a . For the off-diagonals, only adjacent terms $(a_{k+1} - \alpha a_k)^2$ couple a_k and a_{k+1} ; expanding this term contributes a cross derivative $-\alpha/\sigma_\eta^2$, so $Q_0[k, k+1] = -\alpha/\sigma_\eta^2$ and entries at distance ≥ 2 vanish.

For the diagonal at an interior k ($0 < k < K-1$), a_k appears in two transition terms, the left one $(a_k - \alpha a_{k-1})^2$ contributing $1/\sigma_\eta^2$ and the right one $(a_{k+1} - \alpha a_k)^2$ contributing α^2/σ_η^2 , summing to $(1 + \alpha^2)/\sigma_\eta^2$. At the left boundary $k = 0$ the prior contributes $1/\sigma_0^2$ and only the right transition contributes α^2/σ_η^2 ; using stationarity $\sigma_0^2 = \sigma_\eta^2/(1 - \alpha^2)$ the sum simplifies to $1/\sigma_\eta^2$. The right boundary $k = K-1$ is symmetric.

Intuition : tridiagonality is Markovianity, not Gaussianity

The argument used Gaussianity only to identify the precision with the Hessian of $-\log P_0$. The structure of the Hessian – nonzero only on first neighbours – is purely a consequence of the chain factorisation: each transition factor depends on at most two adjacent variables. Any Markov chain with absolutely continuous transitions has a log-density Hessian supported on the path graph; only its *values* depend on the specific transition law.

7.2 Noised covariance and joint score

Proposition 10 (Noised covariance, [Established](#)). $\Sigma_t = \mu^2 \Sigma_0 + \Delta_t I_K$.

Stationary Gaussian AR(1), $K = 20$, $\alpha = 0.9$

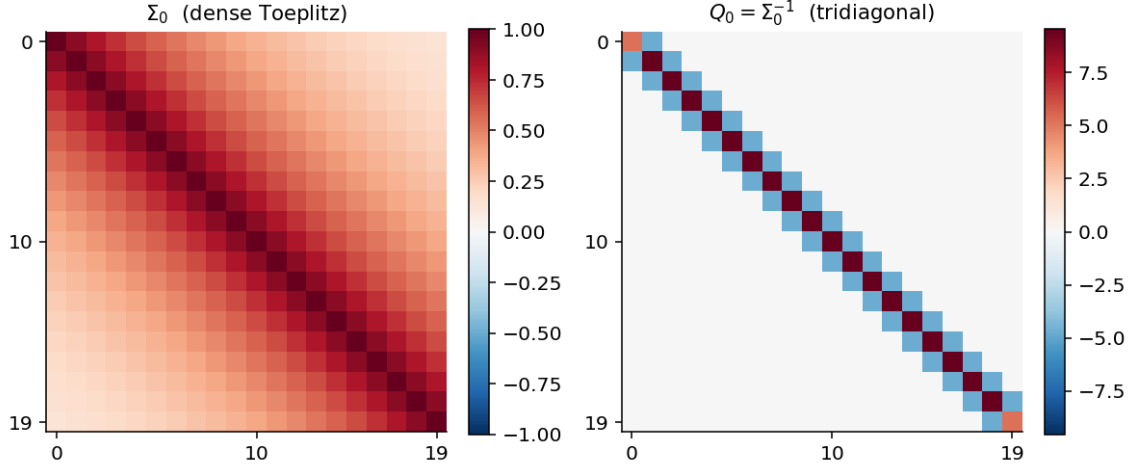


Figure 2: Stationary Gaussian AR(1), $K = 20$, $\alpha = 0.9$. Covariance Σ_0 (left) is dense Toeplitz with geometric off-diagonal decay; the inverse Q_0 (right) is exactly tridiagonal. The only sparse object in the entire Gaussian story is Q_0 , and that sparsity is the algebraic shadow of the chain factorisation.

Derivation : the forward channel is a linear Gaussian map

$x = \mu a + \sqrt{\Delta_t} \xi$ with $a \sim \mathcal{N}(0, \Sigma_0)$ and $\xi \sim \mathcal{N}(0, I)$ independent. Independence of a, ξ gives

$$\text{Cov}(x) = \mu^2 \text{Cov}(a) + \Delta_t \text{Cov}(\xi) = \mu^2 \Sigma_0 + \Delta_t I. \quad (63)$$

Theorem 11 (Affine joint score, [Established]). *For Gaussian AR(1),*

$$S(x, t) = -Q_t(x - \mu \mu_a), \quad Q_t = \Sigma_t^{-1}, \quad \mu_a[k] = \alpha^k \mu_0, \quad (64)$$

which is exactly affine in x at every t .

Derivation 11: from the Gaussian density

$P_t(x) = \mathcal{N}(x; \mu \mu_a, \Sigma_t)$ gives $\log P_t(x) = -\frac{1}{2}(x - \mu \mu_a)^\top Q_t(x - \mu \mu_a) + \text{const}$, whose gradient is $-Q_t(x - \mu \mu_a)$.

Theorem 11 reduces every structural question about the joint score to a question about the matrix Q_t . The remainder of this section describes Q_t in full closed form.

7.3 Three equivalent representations of Q_t

Theorem 12 (Equivalent forms of Q_t , [Established]). *For all $t \geq 0$,*

$$\begin{aligned} Q_t &= \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}, & \gamma_t &:= \frac{\mu^2}{\Delta_t}, & (\text{SNR form}) \\ Q_t &= U \text{diag}(1/(\mu^2 \lambda_i + \Delta_t)) U^\top, & \Sigma_0 &= U \text{diag}(\lambda_i) U^\top, & (\text{spectral form}) \\ Q_t &= e^{2t} Q_0 (I + c_t Q_0)^{-1} = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}, & c_t &:= e^{2t} - 1. & (\text{resolvent form}) \end{aligned}$$

Derivation 12: three rewrites of the same identity

SNR form. Factor Δ_t out of $\Sigma_t = \mu^2 \Sigma_0 + \Delta_t I$ to write $\Sigma_t = \Delta_t(I + (\mu^2/\Delta_t)\Sigma_0) = \Delta_t(I + \gamma_t \Sigma_0)$, and invert.

Spectral form. Decompose $\Sigma_0 = U \text{diag}(\lambda_i) U^\top$. The identity matrix commutes with every basis change, so $\Sigma_t = U \text{diag}(\mu^2 \lambda_i + \Delta_t) U^\top$, and $Q_t = U \text{diag}(1/(\mu^2 \lambda_i + \Delta_t)) U^\top$. The eigenvectors of Σ_0 are exactly the eigenvectors of Σ_t and Q_t ; only the eigenvalues evolve.

Resolvent form. Multiply $\Sigma_t = \mu^2 \Sigma_0 + \Delta_t I$ by Q_0 to get $Q_0 \Sigma_t = \mu^2 I + \Delta_t Q_0$. Since $\mu^2 = e^{-2t} = (1 + c_t)^{-1}$ and $\Delta_t = 1 - \mu^2$, one can verify by direct substitution that $Q_t = e^{2t} Q_0 (I + c_t Q_0)^{-1}$. The Neumann series follows when $\|c_t Q_0\| < 1$, which is the small- t regime where the SNR is high.

Theorem 13 (Spectral preservation, [Established]). *The eigenvectors of Σ_t and Q_t coincide with those of Σ_0 for every $t \geq 0$; only the eigenvalues evolve, as $\lambda_i(t) = \mu^2 \lambda_i + \Delta_t$. In the Σ_0 -eigenbasis the multivariate denoising decouples into K independent scalar denoisers.*

Intuition : locality is a basis-dependent statement

Spectral preservation uses Gaussianity *plus* isotropic noise – no Markov structure is invoked. In the Σ_0 -eigenbasis, Q_t is diagonal at every t – maximally local. In the frame basis, Q_t is tridiagonal only at $t = 0$ – as we will see, the band fills with diffusion time. Both statements describe the same matrix in different coordinates. The frame basis is the operationally relevant one for video data.

7.4 Lifecycle of the precision matrix

The annotated heatmap in Figure 3 shows the actual numerical values of Q_t at six diffusion times for $K = 6$, $\alpha = 0.9$. Three regimes are visible:

- **Tridiagonal** ($t \leq 10^{-5}$): the matrix is indistinguishable from the clean Q_0 ; off-band entries are at floating-point noise ($\sim 10^{-12}$).
- **Banded fill-in** ($10^{-3} \leq t \leq 10^{-1}$): the distance- d entries grow as t^{d-1} , with first neighbours order one, second neighbours order t , third neighbours order t^2 , and so on.
- **Identity floor** ($t \geq 1$): every entry decays exponentially towards the identity at rate e^{-2t} ; at $t = 5$ all off-diagonals are at $\sim 10^{-5}$ and $Q_t \approx I$ to machine precision.

7.4.1 The band-fill theorem (corrected)

The resolvent form (resolvent form) controls how the band fills quantitatively. Powers of a tridiagonal matrix grow linearly in bandwidth – Q_0^m has bandwidth at most m – so the distance- d entry first appears at order t^{d-1} :

Theorem 14 (Band-fill leading order, [Established]). *For the Gaussian AR(1) precision under OU diffusion,*

$$Q_t[i, i + d] = (-1)^{d-1} (2t)^{d-1} (Q_0^d)_{i, i+d} + O(t^d). \quad (65)$$

Frame-basis precision $Q_t = \Sigma_t^{-1}$ at six diffusion times ($K = 6, \alpha = 0.9$). Tridiagonal $\rightarrow$ banded $\rightarrow$ identity-floor.

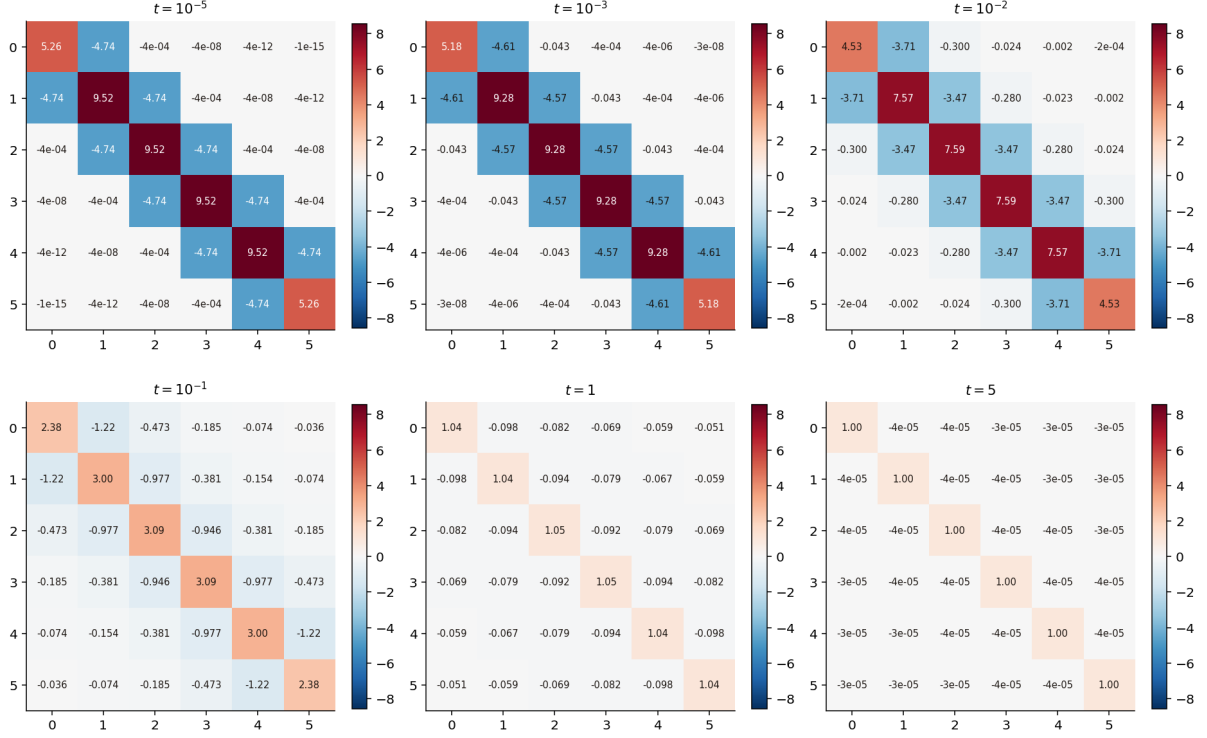


Figure 3: Frame-basis precision Q_t at six diffusion times, $K = 6$, $\alpha = 0.9$, with the actual numerical values written into every cell. Top row: $t = 10^{-5}$ (effectively clean), $t = 10^{-3}$, $t = 10^{-2}$. Bottom row: $t = 0.1$, $t = 1$, $t = 5$. The dense-to-sparse-to-identity lifecycle is visible directly in the entries: tridiagonal structure with ~ 9.5 on the bulk diagonal and -4.7 on the first off-diagonal at $t = 10^{-5}$; weak distance-2 entries of order 10^{-2} appearing already at $t = 10^{-3}$; the matrix densifying through $t = 10^{-1}$; collapsing toward the identity for $t \geq 1$.

Derivation 14: small- t expansion of the resolvent

Substitute $c_t = 2t + 2t^2 + O(t^3)$ and $e^{2t} = 1 + O(t)$ into the Neumann series (resolvent form):

$$Q_t = (1 + O(t)) [Q_0 - c_t Q_0^2 + c_t^2 Q_0^3 - \dots]. \quad (66)$$

The distance- d entry on the left receives contributions only from matrix powers of bandwidth at least d , i.e. Q_0^m with $m \geq d$. The smallest such power is $m = d$, entering the series with prefactor $(-c_t)^{d-1}$. Substituting $c_t = 2t(1 + t + O(t^2))$, $(-c_t)^{d-1} = (-1)^{d-1}(2t)^{d-1}(1 + O(t))^{d-1} = (-1)^{d-1}(2t)^{d-1} + O(t^d)$. Higher m contribute $(-c_t)^{m-1} = O(t^{m-1})$ which is at least $O(t^d)$. Hence (65).

Remark 2 (Correction to a published formula). An earlier internal note (precision_lifecycle_summary.pdf, eq. (12)) prints (65) with an extraneous $1/(d-1)!$ factor. That factor does not follow from the resolvent expansion of equation (10) in the same note, and the audit numerical_audit.py::check_band_fill rejects it: the no-factorial leading coefficient agrees with the empirical $Q_t[i, i+d]/t^{d-1}$ to relative accuracy $< 5\%$ for $d \in \{1, 2, 3, 4, 5\}$ at $\alpha = 0.9$, $K = 25$; the factorial version fails the same test by orders of magnitude (errors $\sim 90\%$ at $d = 3$, $\sim 290\%$ at $d = 4$, $\sim 800\%$ at $d = 5$). The proof sketch in the source PDF itself reads “the leading nonvanishing contribution at distance d comes from

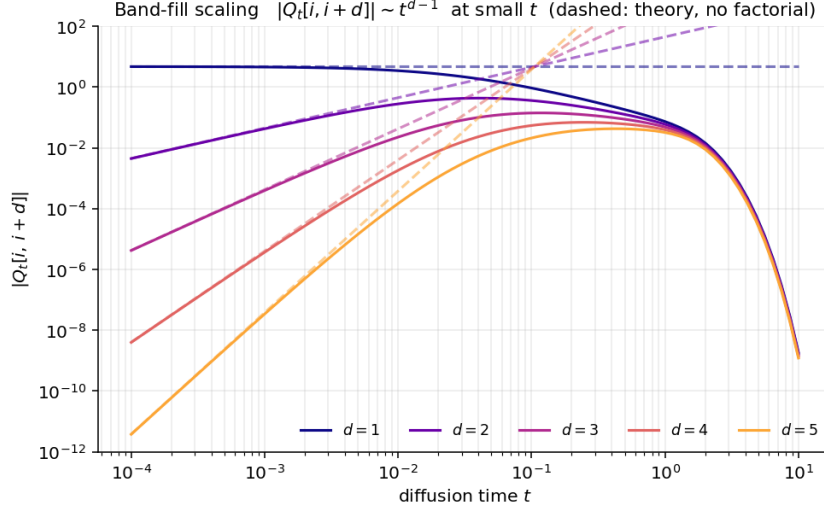


Figure 4: Band-fill scaling. Solid: $|Q_t[i, i+d]|$ for $i = K/2$ and $d = 1, \dots, 5$, computed by direct inversion of Σ_t . Dashed: corrected theoretical leading order (65) (no factorial). The match is exact at small t ; at $t \gtrsim 1$ all curves bend over and decay together as Q_t returns to the identity.

the term $m = d - 1$ ", which is consistent only with the no-factorial formula.

Numerical audit : band-fill verification

`numerical_audit.py::check_band_fill` compares the empirical ratio $Q_t[i, i+d]/t^{d-1}$ to the predicted leading coefficient $(-1)^{d-1}2^{d-1}(Q_0^d)_{i,i+d}$ for each $d \in \{1, \dots, 5\}$, at the smallest t consistent with floating-point recovery of the off-band entries. All five passes at relative tolerance 5%. The figure `fig04_band_fill.png` (Figure 4) plots $|Q_t[i, i+d]|$ on log-log axes against the corrected reference and confirms the match across the small- t regime.

7.4.2 Asymptotic exponential return to identity

Theorem 15 (Large- t asymptotic, [Established]). As $t \rightarrow \infty$,

$$Q_t = I - e^{-2t}(\Sigma_0 - I) + e^{-4t}(\Sigma_0 - I)^2 + O(e^{-6t}). \quad (67)$$

Derivation 15: Neumann at the identity

Write $\Sigma_t = I + e^{-2t}(\Sigma_0 - I)$ (substituting $\mu^2 = e^{-2t}$ and $\Delta_t = 1 - e^{-2t}$). Apply $(I + B)^{-1} = I - B + B^2 - \dots$ with $B = e^{-2t}(\Sigma_0 - I)$. The norm $\|B\| = e^{-2t}\|\Sigma_0 - I\|$ is below 1 for t large enough; the series converges and the first three terms are as stated.

7.5 The full-chain BP score recovers the matrix score

Section 6 derived an $O(K)$ algorithm for the joint score by message passing. For Gaussian AR(1), the resulting score must equal the closed-form expression of Theorem 11. The audit confirms this to machine precision over 1600 configurations.

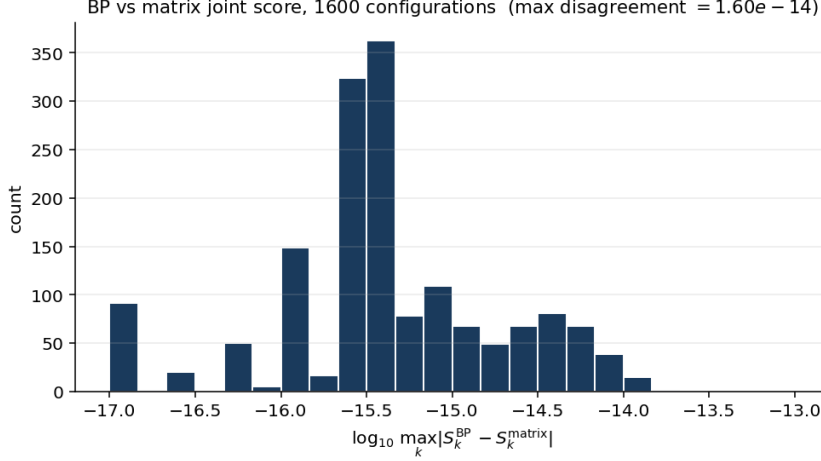


Figure 5: Histogram of $\log_{10} \max_k |S_k^{\text{BP}} - S_k^{\text{matrix}}|$ over 1600 audit configurations. The bulk of residuals concentrates at $\log_{10} \approx -15.5$ (machine-eps level for the score’s magnitude); the leftmost floor bin is the clip at 10^{-17} for configurations whose BP and matrix scores agree exactly to all displayed digits.

Numerical audit : BP vs matrix posterior, 80 configurations $\times$ 20 random x

`numerical_audit.py::check_bp_vs_matrix` sweeps $K \in \{2, 3, 5, 8, 16\}$, $\alpha \in \{0.2, 0.5, 0.9, -0.4\}$, $t \in \{0.05, 0.3, 1.0, 3.0\}$, with 20 random $x \sim \mathcal{N}(0, I)$ per configuration. Maximum disagreements:

$$\begin{aligned} \max |m_k^{\text{BP}} - m_k^{\text{matrix}}| &= 1.33 \times 10^{-15}, \\ \max |v_k^{\text{BP}} - \Sigma_{\text{post}}[k, k]| &= 3.89 \times 10^{-15}, \\ \max |S_k^{\text{BP}} - S_k^{\text{matrix}}| &= 1.24 \times 10^{-14}. \end{aligned}$$

All three are at the level of double-precision rounding accumulated through the recursion (K products of three Gaussians, each introducing at worst one ULP).

8 Act II — Laplace warm-up at $K = 1$

For $K = 1$ there is no chain and no Markov structure; the model is the smallest case in which Tweedie’s identity has to confront a non-Gaussian prior. The clean variable is $a \sim \text{Laplace}(0, b)$, $p_0(a) = (1/2b)e^{-|a|/b}$, $\text{Var}(a) = 2b^2$. The OU step yields $p_t(x) = \int p_0(a) \mathcal{N}(x; \mu a, \Delta_t) da$.

Programme for the Laplace case (Garnier-Brun discussion). The current section and the next address three threads opened in discussion with Jérôme:

1. express the joint score $S(x, t)$ using the Markov structure of the model;
2. rewrite the score in terms of *relative* (innovation) variables, making the dependence on the AR(1) shift explicit;
3. exploit a Fourier representation in which the noised marginal satisfies a *linear* ODE, recovering the $K = 1$ building block as a screened Poisson equation.

Sections 8.2 and 8.3 address threads 3 and (briefly) the resulting curvature observation for $K = 1$; section 9 addresses threads 1, 2, 3 for $K \geq 2$.

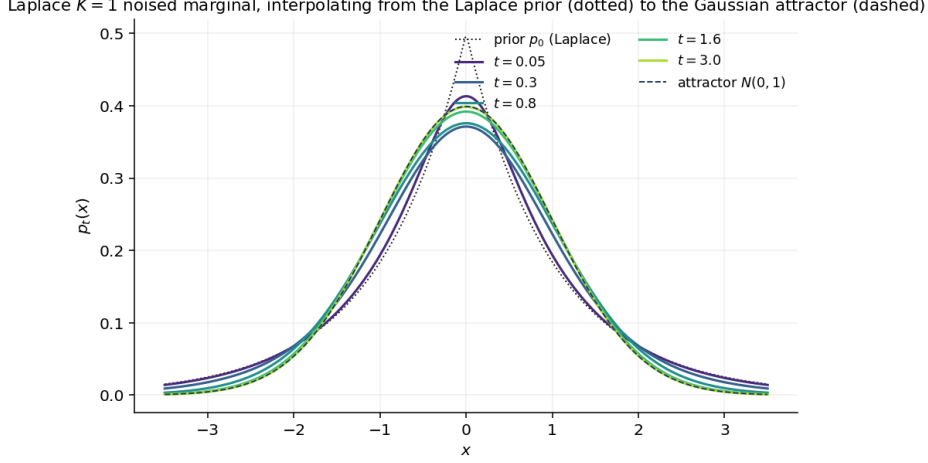


Figure 6: Laplace $K = 1$ noised marginal $p_t^{(1)}(x)$ ($b = 1$) at increasing diffusion times. The dotted curve is the clean Laplace prior $p_0(x) = \frac{1}{2}e^{-|x|}$; the dashed curve is the Gaussian attractor $\mathcal{N}(0, 1)$. The family interpolates monotonically from the kinked Laplace shape at $t \rightarrow 0$ to the smooth Gaussian at $t \rightarrow \infty$.

Remark 3 (On the closed form). The original note `session_summary_K1_warmup.pdf` reduces $p_t(x)$ to a sum of two truncated-Gaussian terms (§ 3.3) using a Mills-ratio asymptotic (Appendix B). We *omit* that derivation: the closed form is convenient for board-work, but every quantitative claim below is established by 1D quadrature in `code/laplace_score.py`, which is independent of the truncated-Gaussian rewrite.

Computation : limits of the noised marginal

$t \rightarrow 0$. As $\mu \rightarrow 1$ and $\Delta_t \rightarrow 0$, the Gaussian kernel $(2\pi\Delta_t)^{-1/2} \exp[-(x - \mu a)^2/2\Delta_t]$ becomes a Dirac at $a = x/\mu \rightarrow x$. The sifting property gives $p_0(x)$: the clean Laplace density is recovered.

$t \rightarrow \infty$. As $\mu \rightarrow 0$ and $\Delta_t \rightarrow 1$, the Gaussian kernel becomes a -independent and factors out, leaving $\int p_0(a) da = 1$; hence $p_\infty(x) = (2\pi)^{-1/2} \exp(-x^2/2)$, the standard normal attractor of the OU process.

8.1 The joint score acquires nonlinearity

By Tweedie (Theorem 1), $S(x, t) = (\mu \mathbb{E}[a | x] - x)/\Delta_t$; the score's x -dependence is exactly that of the Bayesian denoiser.

Intuition : where the nonlinearity lives

For Gaussian p_0 , $\mathbb{E}[a | x]$ is affine in x (Gaussian conditioning), so the score is affine. For Laplace p_0 , the conditional expectation is a smooth nonlinear interpolation between the two limits below, and the score inherits the nonlinearity.

Computation : limits of the joint score

$t \rightarrow 0$. Since $p_t(x) \rightarrow p_0(x) = (1/2b)e^{-|x|/b}$ in the sense above, $\partial_x \log p_t(x) \rightarrow -\text{sign}(x)/b$. The score per se diverges with Δ_t^{-1} ; the rescaled score $\Delta_t S(x, t) \rightarrow -\text{sign}(x)/b$ (up to the

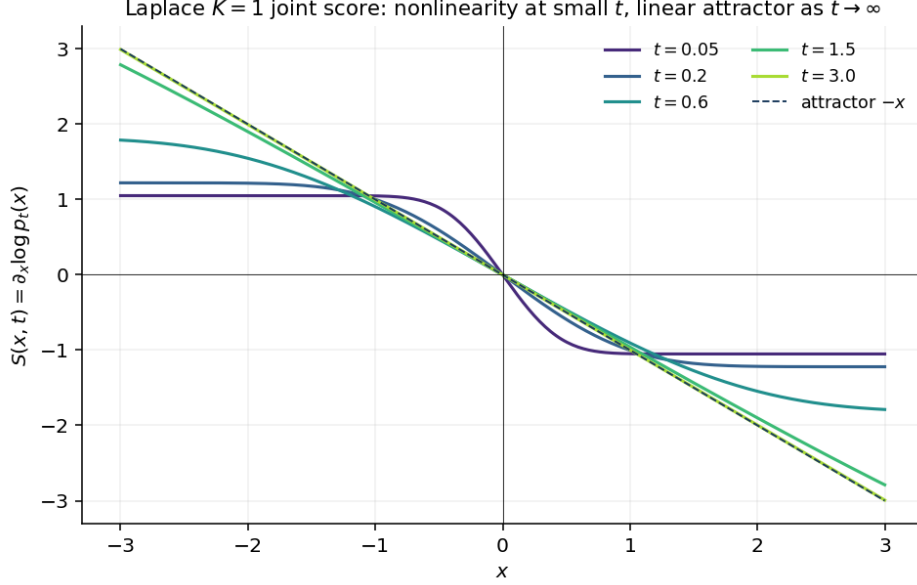


Figure 7: Laplace $K = 1$ joint score $S(x, t) = \partial_x \log p_t(x)$ ($b = 1$) at the same diffusion times as Figure 6; dashed line is the Gaussian attractor $-x$. At small t the score saturates to a sign-like profile of magnitude $\sim 1/b$; the nonlinearity sits in the kink near $x = 0$ and progressively flattens out, recovering linearity as $t \rightarrow \infty$.

factor b in our scaling). Figure 8 (left) shows this convergence for $b = 1$. $t \rightarrow \infty$. As $p_t(x) \rightarrow \mathcal{N}(0, 1)$, the score $\partial_x \log p_t(x) \rightarrow -x$: the Gaussian attractor of the OU process. Figure 8 (right) shows this convergence.

Numerical audit : Tweedie versus finite difference

`numerical_audit.py::check_laplace_K1_score_fd` computes the central finite difference of $\log p_t$ at step $h = 5 \times 10^{-4}$ for $(b, t, x) \in \{0.5, 1\} \times \{0.1, 0.4, 1, 2\} \times \{-1, -0.3, 0.2, 0.7, 1.5\}$. The maximum disagreement against Tweedie is 1.19×10^{-7} , within the $O(h^2)$ truncation floor of the finite difference. The Gaussian-attractor limit at $t = 6$ is verified to 7.4×10^{-6} .

8.2 Fourier representation: convolution becomes a product

The forward channel $y = \mu a + \sqrt{\Delta_t} \xi$ writes the noised variable as the sum of two *independent* random variables: the rescaled clean signal μa and the channel noise $\sqrt{\Delta_t} \xi$. At the level of densities this independence appears as a convolution, $p_t^{(1)} = p_{\mu a} \star G_{\Delta_t}$, where $p_{\mu a}$ is the density of μa and G_{Δ_t} is the Gaussian of variance Δ_t . Convolutions are hard to manipulate directly, but the Fourier transform turns them into ordinary products: the characteristic function of a sum of independent variables is the product of their characteristic functions. That is the structural fact we now exploit.

Theorem 16 (Characteristic-function factorisation, [Established]). *For $a \sim \text{Laplace}(0, b)$, $\xi \sim \mathcal{N}(0, 1)$ independent, and $y = \mu a + \sqrt{\Delta_t} \xi$,*

$$\hat{p}_t^{(1)}(k) = \underbrace{\frac{1}{1 + b^2 \mu^2 k^2}}_{\text{c.f. of } \mu a} \cdot \underbrace{e^{-\Delta_t k^2 / 2}}_{\text{c.f. of } \sqrt{\Delta_t} \xi}. \quad (68)$$

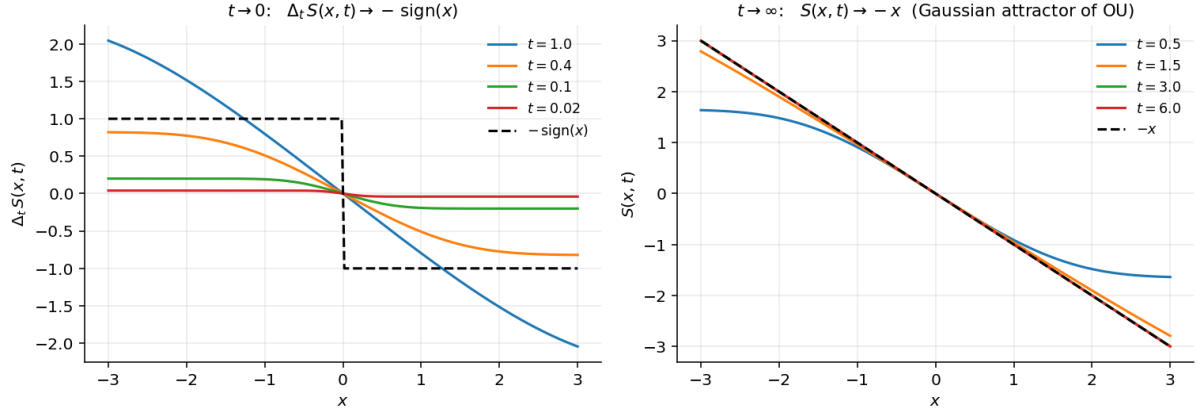


Figure 8: Limits of the Laplace $K = 1$ joint score ($b = 1$). Left: rescaled score $\Delta_t S(x, t)$ for decreasing t ; the family converges pointwise to $-\text{sign}(x)$. Right: unscaled score for increasing t ; the family converges pointwise to $-x$, the Gaussian attractor of the OU process.

Derivation 16: convolution $\leftrightarrow$ product

Step 1 — the density of y is a convolution. Because μa and $\sqrt{\Delta_t} \xi$ are independent, the density of their sum is the convolution of their individual densities:

$$p_t^{(1)}(y) = \int_{-\infty}^{+\infty} p_{\mu a}(y - z) G_{\Delta_t}(z) dz = (p_{\mu a} \star G_{\Delta_t})(y). \quad (69)$$

Step 2 — Fourier turns convolution into product. The convolution theorem states that the Fourier transform of a convolution is the pointwise product of the individual transforms. Taking the characteristic function (Fourier transform with conjugate sign convention) of both sides,

$$\hat{p}_t^{(1)}(k) = \hat{p}_{\mu a}(k) \cdot \hat{G}_{\Delta_t}(k). \quad (70)$$

Independence of a and ξ is precisely what allows the joint expectation $\mathbb{E}[e^{iky}] = \mathbb{E}[e^{ik(\mu a + \sqrt{\Delta_t} \xi)}]$ to split into $\mathbb{E}[e^{ik\mu a}] \mathbb{E}[e^{ik\sqrt{\Delta_t} \xi}]$, which is (70) written out.

Step 3 — the two factors are standard characteristic functions. For $a \sim \text{Laplace}(0, b)$ with density $\frac{1}{2b} e^{-|a|/b}$, a direct integral gives $\hat{p}_a(k) = (1 + b^2 k^2)^{-1}$; rescaling by μ sends $k \mapsto \mu k$, so $\hat{p}_{\mu a}(k) = (1 + b^2 \mu^2 k^2)^{-1}$. For $\xi \sim \mathcal{N}(0, 1)$, $\sqrt{\Delta_t} \xi \sim \mathcal{N}(0, \Delta_t)$ and $\hat{G}_{\Delta_t}(k) = e^{-\Delta_t k^2/2}$. Multiplying the two factors gives (68).

Intuition : why working in Fourier is the right move

In real space $p_t^{(1)}$ is the convolution of a rough function (Laplace, with a kink at 0) with a smooth Gaussian; manipulating the convolution directly forces one to confront the kink at every step. In Fourier space the two factors of (68) sit *side by side as a pointwise product*: any operation we apply on the left ends up multiplying both factors. This is exactly what makes the linear ODE below possible.

Theorem 17 (Screened Poisson equation, [\[Established\]](#)). *The noised marginal $p_t^{(1)}(y)$ satisfies the linear, constant-coefficient ODE*

$$(1 - b^2 \mu^2 \partial_y^2) p_t^{(1)}(y) = G_{\Delta_t}(y), \quad G_{\Delta_t}(y) := \frac{1}{\sqrt{2\pi\Delta_t}} e^{-y^2/(2\Delta_t)}. \quad (71)$$

Derivation 17: clearing the denominator and undoing Fourier

Step 1 — multiply both sides of (68) by $1 + b^2\mu^2k^2$. This removes the rational factor on the right:

$$(1 + b^2\mu^2k^2) \widehat{p}_t^{(1)}(k) = e^{-\Delta_t k^2/2}. \quad (72)$$

The right-hand side is the characteristic function of the Gaussian G_{Δ_t} , so its inverse Fourier transform is $G_{\Delta_t}(y)$ itself.

Step 2 — the differentiation rule $k^2 \widehat{f} \leftrightarrow -\partial_y^2 f$. For any density f decaying at infinity, $\widehat{(\partial_y f)}(k) = ik \widehat{f}(k)$; applying it twice gives $\widehat{(\partial_y^2 f)}(k) = (ik)^2 \widehat{f}(k) = -k^2 \widehat{f}(k)$. Inverting,

$$k^2 \widehat{f}(k) \xleftrightarrow{\text{Fourier}} -\partial_y^2 f(y). \quad (73)$$

Second-order differentiation in real space corresponds to multiplication by $-k^2$ in Fourier space. The minus sign is the single fact that produces a sign change between the rational factor $(1 + b^2\mu^2k^2)$ in Fourier and the differential operator $(1 - b^2\mu^2\partial_y^2)$ in real space.

Step 3 — inverse-transform (72) term by term. The constant 1 on the left is the identity operator. The $b^2\mu^2k^2$ term, by (73), corresponds to $-b^2\mu^2\partial_y^2$. The Fourier inverse of the right-hand side is $G_{\Delta_t}(y)$. Combining,

$$(1 - b^2\mu^2\partial_y^2) p_t^{(1)}(y) = G_{\Delta_t}(y), \quad (74)$$

which is (71). □

Intuition : a linear ODE with a Gaussian source

(71) says that the noised Laplace marginal is the solution of a second-order, linear, constant-coefficient ODE driven by a Gaussian source. This is the structural advantage that the Fourier representation extracts: a convolution of two distinct densities in real space has been turned into the action of a single local differential operator on $p_t^{(1)}$. The operator $(1 - b^2\mu^2\partial_y^2)$ is the inverse of the rational factor $(1 + b^2\mu^2k^2)^{-1}$ in (68); the duality $k^2 \leftrightarrow -\partial_y^2$ is what allows it to be written in this clean form. The further analysis of the ODE (71) – its Green’s function, the interpretation of $b\mu$ as a screening length, the regimes $t \rightarrow 0$ and $t \rightarrow \infty$ – is a programme objective in its own right and is not pursued in this note; (71) is the structural statement we wanted.

Numerical audit : characteristic-function identity

`numerical_audit.py::check_laplace_K1_fourier` verifies the closed-form characteristic function (68) numerically by comparing it against a quadrature of the density against e^{iky} on $b \in \{0.5, 1\}$, $t \in \{0.2, 0.6, 1.4\}$, $k \in \{0.3, 1.0, 2.5\}$; maximum disagreement 2.88×10^{-6} at the floor of adaptive quadrature on an oscillatory unbounded-domain integrand.

8.3 The curvature is x -dependent

Tweedie’s identity (Theorem 1) gives, by direct differentiation of the score with respect to x ,

$$H_t(x) := -\partial_x^2 \log p_t(x) = \frac{1}{\Delta_t} \left(1 - \frac{\mu^2}{\Delta_t} \text{Var}[a \mid x] \right). \quad (75)$$

For Gaussian p_0 , $\text{Var}[a \mid x]$ is a constant in x (Gaussian conditioning has x -independent posterior variance), so H_t is a constant – the Hessian of an affine score. For Laplace p_0 , the conditional variance is x -dependent, so $H_t(x)$ is a nontrivial *field* on $\mathbb{R}$. We do not develop the full analysis of $H_t(x)$ here – it is a programme objective in its own right, with quadrature-tractable expressions and bump-location results that the dedicated `laplace_ar1_audit/` package treats in detail. The point we want to record is simply that the breakdown of *Hessian constancy* is the second-order analogue of the breakdown of *score linearity*, and both follow from the same source: a non-Gaussian prior turns $\mathbb{E}[a \mid x]$ into a nonlinear function of x .

9 Act III — Laplace AR(1) at $K \geq 2$

The Laplace AR(1) chain has joint density

$$P_0(a) = \frac{1}{(2b)^K} \exp\left(-\frac{1}{b} \sum_{k=0}^{K-1} |a_k - \alpha a_{k-1}|\right), \quad a_{-1} := 0. \quad (76)$$

The likelihood factorises across frames as in the Gaussian case, so the posterior $p(a \mid x)$ inherits the same chain factor graph. Belief propagation is therefore still *exact in principle*, but the messages no longer remain Gaussian; computation requires either 1D quadrature or a tractable analytic representation. This section addresses the three threads opened in section 8: the Markov-structured form of the score, its rewrite in relative coordinates, and the Fourier programme that promotes the $K = 1$ screened Poisson identity (71) to a recursive linear structure.

9.1 The Markov-structured form of the joint score

Theorem 18 (Joint score from chain BP, [Established]). *For any chain prior P_0 (in particular Laplace AR(1)) and the OU likelihood, the joint score $S(x, t) \in \mathbb{R}^K$ is given component-wise by*

$$S_k(x, t) = \frac{\mu \mathbb{E}[a_k \mid x] - x_k}{\Delta_t}, \quad \mathbb{E}[a_k \mid x] = \frac{\int a_k m_k(a_k) p_t(x_k \mid a_k) b_k(a_k) da_k}{\int m_k(a_k) p_t(x_k \mid a_k) b_k(a_k) da_k}, \quad (77)$$

where $m_k(a_k) := \mu_{\rightarrow k}(a_k)$ is the forward message and $b_k(a_k) := \mu_{\leftarrow k}(a_k)$ is the backward message of section 6, Convention A.

Derivation 18: Tweedie + sum-product

Tweedie (Theorem 1) gives $S_k(x, t) = (\mu \mathbb{E}[a_k \mid x] - x_k)/\Delta_t$. The conditional expectation reduces to a single 1D integral via the marginal posterior $p(a_k \mid x) \propto m_k(a_k) p_t(x_k \mid a_k) b_k(a_k)$ (Proposition 7). The proportionality constant cancels between the numerator and the denominator of the conditional expectation, leaving the ratio of two one-dimensional integrals as in (77).

Intuition : this answers Jérôme’s first thread

(77) is the explicit answer to “express the score using the Markov structure of the model.” The cost is no longer a K -dimensional integral of the joint posterior; it is two one-dimensional message recursions plus one one-dimensional integral per frame for the conditional expectation. For Gaussian P_0 , the messages are Gaussian and the integrals collapse to arithmetic on (m, v) pairs (section 6); for Laplace P_0 , the messages are no longer Gaussian, but the forward message still admits a clean closed form (Proposition 19, below) that recycles the $K = 1$ screened-Poisson kernel.

9.2 The boundary message has a clean closed form

The recursion (55) of Convention A starts from the uninformative $\mu_{\leftarrow K-1} \equiv 1$. Its first iteration – the step that combines the AR(1) transition with the local evidence of a single frame against the trivial incoming message – has a clean closed form.

Proposition 19 (One-step Laplace integral, [Established]). *For the Laplace AR(1) transition $M(a' | a) = (1/2b) e^{-|a' - \alpha a|/b}$ and the OU local evidence $p_t(x' | a') = \mathcal{N}(x'; \mu a', \Delta_t)$,*

$$\int_{\mathbb{R}} M(a' | a) p_t(x' | a') da' = p_t^{(1)}(x' - \mu \alpha a), \quad (78)$$

viewed as a function of a , with $p_t^{(1)}$ the universal $K = 1$ Laplace noised marginal of section 8. In particular, the first backward step of Convention A (with $\mu_{\leftarrow k} \equiv 1$) reads $\mu_{\leftarrow k-1}(a_{k-1}) = p_t^{(1)}(x_k - \mu \alpha a_{k-1})$.

Derivation 19: change of variable to the innovation

Introduce the innovation variable $r := a' - \alpha a$ (with a held fixed). The transition density in this variable is $M(a' | a) = (1/2b) e^{-|r|/b}$, i.e. the bare Laplace prior of section 8 evaluated at r . The Gaussian likelihood at $a' = \alpha a + r$ is $\mathcal{N}(x'; \mu(\alpha a + r), \Delta_t)$, which is just the OU kernel applied to the shifted argument $x' - \mu \alpha a$ at innovation r . Marginalising r out,

$$\int \frac{1}{2b} e^{-|r|/b} \mathcal{N}(x'; \mu(\alpha a + r), \Delta_t) dr = \int p_0(r) \mathcal{N}(x' - \mu \alpha a; \mu r, \Delta_t) dr = p_t^{(1)}(x' - \mu \alpha a), \quad (79)$$

using the definition of the $K = 1$ noised marginal. The shift $\mu \alpha a$ is precisely what frame a predicts for the next frame's noised observation under the AR(1) dynamics, scaled by the channel μ .

9.3 General (non-boundary) messages have no clean form

Remark 4 (Where the clean form breaks down, [Established]structurally). (78) holds because the integrand involves *exactly one* Laplace kink (from M) multiplying *exactly one* Gaussian (from $p_t(x' | a')$). Any subsequent step of either BP recursion — in either direction — breaks both conditions: the message coming in carries the propagated structure of several earlier frames, and is itself a sum of products of Laplace kinks and Gaussian factors. The integrand at a general step is then a product of *two or more* absolute-value kinks with Gaussian factors, which on each sign-quadrant reduces to a half-line Gaussian but globally gives a sum of error functions whose argument depends on the conditioning variable in a non-affine way. The closed form of the boundary step does not survive iteration; this is the technical content of the asymmetry between the Laplace and the Gaussian benchmarks (in the Gaussian case every step is Gaussian-times-Gaussian and the clean form propagates).

9.4 Innovation coordinates: prior factorises, likelihood densifies

The natural rewrite of the Laplace AR(1) prior is in terms of *innovation* variables that absorb the AR(1) shift. The result makes the duality between prior and likelihood very explicit.

Proposition 20 (Innovation factorisation, [Established]). *The triangular change of variables*

$$r_0 = a_0, \quad r_k = a_k - \alpha a_{k-1} \quad \text{for } k \geq 1, \quad (80)$$

has unit Jacobian and decouples the prior into iid Laplace coordinates:

$$P_0(r) = \frac{1}{(2b)^K} \prod_{k=0}^{K-1} e^{-|r_k|/b}. \quad (81)$$

Writing $a = Lr$ with L unit lower-triangular ($L_{ij} = \alpha^{i-j} \mathbf{1}_{i \geq j}$), the OU likelihood becomes

$$p(x | r) \propto \exp\left(-\frac{\mu^2}{2\Delta_t} r^\top L^\top L r + \frac{\mu}{\Delta_t} x^\top L r\right), \quad (82)$$

and the matrix $L^\top L$ is dense.

Derivation 20: change of variables in detail

Step 1 — the inverse map is lower-triangular. The defining relations (80) read off the entries of r from a : $r_0 = a_0$ and $r_k = a_k - \alpha a_{k-1}$. Reading them the other way around, $a_k = \alpha a_{k-1} + r_k$, and unrolling the recursion starting from $a_0 = r_0$,

$$a_k = \sum_{j=0}^k \alpha^{k-j} r_j, \quad k = 0, \dots, K-1. \quad (83)$$

This is $a = Lr$ with $L_{ij} = \alpha^{i-j} \mathbf{1}_{i \geq j}$; the diagonal is $L_{ii} = 1$ so L is unit lower-triangular and $\det L = 1$.

Step 2 — the Jacobian of the change of variables is one. The Jacobian determinant of the linear map $a = Lr$ is $|\det L| = 1$, so the Lebesgue measure is invariant: $da = dr$. Hence pushing forward any density amounts to substituting $a = Lr$ without an extra factor.

Step 3 — the prior factor by factor. The Laplace AR(1) prior is $P_0(a) = (2b)^{-K} \exp(-\frac{1}{b} \sum_k |a_k - \alpha a_{k-1}|)$ with $a_{-1} := 0$. By (80), each kink $|a_k - \alpha a_{k-1}|$ is simply $|r_k|$; the $k = 0$ term $|a_0 - 0|$ is $|r_0|$. Substituting gives (81): a product of K iid Laplace densities. The prior in r -coordinates is therefore as decoupled as it can possibly be.

Step 4 — the likelihood in r -coordinates. The OU likelihood $p(x | a) \propto \exp(-\frac{1}{2\Delta_t} \sum_k (x_k - \mu a_k)^2)$ is quadratic in a , so substituting $a = Lr$ gives

$$\begin{aligned} \sum_{k=0}^{K-1} (x_k - \mu a_k)^2 &= \|x - \mu Lr\|^2 \\ &= x^\top x - 2\mu x^\top Lr + \mu^2 (Lr)^\top (Lr) \\ &= \mu^2 r^\top L^\top L r - 2\mu x^\top Lr + \text{const}_r. \end{aligned} \quad (84)$$

Dividing by $2\Delta_t$ and exponentiating gives (82). The cost of decoupling the prior is the cross term $r^\top L^\top L r$: $L^\top L$ is dense because the unit-triangular L has all entries $L_{ij} = \alpha^{i-j}$ in its lower triangle, so $(L^\top L)_{ij} = \sum_m L_{mi} L_{mj}$ sums geometric series in α and is non-zero for every pair (i, j) .

Intuition : the AR(1) coupling cannot disappear

The change to innovation coordinates is the cleanest possible rewriting of the prior: it factorises completely into iid Laplace variables. But the original AR(1) coupling did not vanish; it migrated. In original coordinates the prior carried the chain coupling and the likelihood was diagonal; in innovation coordinates the prior is diagonal and the likelihood carries the coupling, through the dense $L^\top L$. There is no choice of basis in which both

prior and likelihood are diagonal simultaneously. The non-Gaussian counterpart of the Gaussian observation in §7 — that Σ_t^{-1} is tridiagonal in the frame basis and diagonal in the spectral basis, but never both — is that the AR(1) shift has to live somewhere.

9.5 What the structure buys (and what is open)

[Established] Markovianity reduces the global K -dimensional posterior expectation to two one-dimensional message recursions, exactly as in the Gaussian case; computational locality survives. The forward message is the universal $K = 1$ noised density at a shifted argument (Proposition 19), so the Act II analysis recycles as a building block.

[Open] A quadrature-tractable representation of the backward message at $K \geq 2$, or a controlled small- α expansion, would close the gap.

[Open] A precise, basis-independent statement of which *locality* of the joint score survives the non-Gaussian extension. The candidates are sparsity of the position-dependent Hessian $H_t(x) = -\nabla_x^2 \log P_t(x)$ in some basis, low-rank corrections to a tridiagonal core, or a perturbative expansion in α or in the strength of the non-Gaussianity. The earlier project’s quick-resolution low-rank conjecture (`conjecture15_test_note.md`) is regime-dependent and pending HPC verification; we do not commit to it here.

10 Numerical audit summary

The driver `code/numerical_audit.py` runs 53 closed-form-vs-reference checks. The script exits non-zero if any check fails. **As of compile date, 53/53 checks pass.**

Numerical audit : full check inventory

Identity	max error	tolerance
Clean precision $Q_0 = \Sigma_0^{-1}$ (12 (K, α) configs)	1.78×10^{-14}	10^{-12}
Q_0 strictly tridiagonal, same configs	0	exact
$\Sigma_t Q_t = I$, 5 diffusion times	8.88×10^{-16}	10^{-10}
Q_t direct vs spectral form, 5 times	2.22×10^{-14}	10^{-10}
U diagonalises Σ_0 , 5 times	1.19×10^{-15}	10^{-10}
Band-fill leading coefficient, $d \in \{1, \dots, 5\}$ (corrected formula (65))	2.7×10^{-2} (rel.)	0.05 (rel.)
$\ Q_t - I\ _F$ vs large- t prediction, $t \in \{3, 6, 10\}$	2.08×10^{-2} (rel.)	$\propto e^{-2t}$
BP vs matrix posterior mean (80 configs)	1.33×10^{-15}	10^{-10}
BP vs matrix posterior variance (80 configs)	3.89×10^{-15}	10^{-10}
BP vs matrix joint score (80 configs)	1.24×10^{-14}	10^{-10}
Tweedie matrix score self-consistency (Gaussian AR(1))	2.66×10^{-15}	10^{-12}
Laplace $K = 1$ score: Tweedie vs central FD	1.19×10^{-7}	5×10^{-7}
Laplace $K = 1$ score $\rightarrow -x$ at $t = 6$	7.37×10^{-6}	10^{-3}

11 Closing argument: the factor graph as inductive bias

The previous sections established three connected facts. First, Tweedie’s identity (Theorem 1) collapses the score to the Bayesian denoiser $\mathbb{E}[a \mid x]$, so the prior P_0 enters the score *only* through this conditional expectation. Second, the chain prior factorises *and* the OU likelihood factorises, so the posterior $P(a \mid x)$ is a tree-structured graphical model. Third, on a tree, sum-product

is exact and runs in $O(K)$, identifying with the Kalman filter and the RTS smoother in the Gaussian case. We close by drawing the architectural consequence.

11.1 The intuition, made precise

Intuition : structure-aware versus universal-approximator priors

The score-matching paradigm trains a neural network $s_\theta(x, t)$ to approximate $\nabla_x \log P_t(x)$. The network is a universal approximator: in absence of architectural constraints, the prior over functions s_θ it implements has *very high entropy* – almost any reasonable smooth field can be represented for some θ . The training data has to do all the work of locating the right function in this enormous space.

For our problem, P_0 is a stationary AR(1) chain. By Theorem 1, the score is determined by $\mathbb{E}[a \mid x]$. By Proposition 4, $\mathbb{E}[a_k \mid x]$ decomposes through the chain factor graph. The forward and backward Kalman recursions deliver $\mathbb{E}[a_k \mid x]$ in $O(K)$ operations, with no training required (the recursion coefficients are functions of α, σ_η, t , all known). Equivalently: the chain structure of the prior *is* a much more informative architectural prior over score fields than “unconstrained smooth”, and using it collapses the model class to one that is exact for the assumed generative model and $O(K)$ to evaluate.

11.2 The convolutional analogy

A standard fully-connected MLP applied to images is a universal approximator over functions of pixel vectors: it does not know that the input is an image, that pixels have spatial neighbours, or that visual patterns are typically translation-equivariant. Convolutional networks bake exactly these facts into the architecture: local filters, weight sharing, pooling. The result is a smaller hypothesis class that contains the right answers for visual data, and that is trainable from far fewer samples than the unstructured MLP would require. The convolutional architecture *is* an informative prior; the training data is no longer asked to discover locality and translation-equivariance, only to fit the residual.

Our setting is analogous, with locality of *visual* pattern replaced by locality of *temporal* dependence. The information we have about the data-generating process – that it is a stationary Markov chain under known iid Gaussian corruption – is exactly the information that converts a generic score-field hypothesis class into the sum-product algorithm on the chain factor graph. Implementing the score model as that algorithm:

- **Reduces the parameter count** from a generic neural score field (millions of weights) to the AR(1) transition parameters $(\alpha, \sigma_\eta, p_0)$ plus, in the non-Gaussian case, a finite-dimensional parametrisation of the messages.
- **Reduces the inference cost** from a forward pass through a deep network to two $O(K)$ sequential passes whose total scalar arithmetic is comparable to a few attention layers but with provable exactness in the Gaussian case.
- **Provides a controlled non-Gaussian extension.** Section 9 shows the chain factor graph survives non-Gaussian priors; the only obstruction is the analytic representation of non-Gaussian messages, which is the right hard problem to attack next.

Intuition : the programme objective in one line

Replace the universal-approximator score model with a factor-graph score model whose architecture mirrors the Markov structure of the prior. Train the latter, when training is needed, only on the residual that is not already captured by the chain factorisation.

11.3 Open programme directions

1. Quadrature-tractable backward message at $K \geq 2$ for the Laplace prior, possibly via a Fourier or screened-Poisson representation borrowed from the $K = 1$ analysis.
2. Explicit connection between the joint propagator $P_t : S_{k-1}(x, t) \mapsto S_k(x, t)$ and the Kalman gain in the Gaussian case; quantification of when an approximate P_t extracted from the exact Gaussian solution is a useful inductive bias for non-Gaussian chains. The dynamical-regime picture of [3] suggests this question is sharpest near the speciation transition.
3. Empirical comparison: on a synthetic AR(1)-with-Laplace-noise benchmark, train a generic temporal U-Net score model and compare sample quality, compute, and parameter count against a factor-graph-structured score with the same training budget.

Reproducibility

The `code/` directory contains five modules:

ar1_utils.py

Stationary AR(1) covariance and tridiagonal precision; OU forward channel; Σ_t and Q_t (direct and spectral); affine joint score; matrix Gaussian posterior used by the audit.

bp_score.py

Belief propagation under Convention A; forward (Kalman filter) and backward (RTS smoother) recursions; posterior mean and joint score via Tweedie. Handles the uninformative initial backward message exactly through $v = +\infty$.

laplace_score.py

$K = 1$ Laplace warm-up: noised marginal, posterior mean, and joint score by 1D quadrature; Gaussian $K = 1$ reference for side-by-side comparisons.

numerical_audit.py

53 independent checks against analytic, matrix, finite-difference, and quadrature references; exits non-zero if any check fails.

generate_figures.py

Reproduces all 8 PNGs of this note deterministically.

To reproduce:

```
cd Score_Diffusion/Research/unified_document/code
python3 numerical_audit.py      # MUST report 53/53 PASSED
python3 generate_figures.py     # writes ../figures/fig0{1..8}*.png
cd ..
tectonic main.tex              # preferred: single-binary engine
# or: pdflatex main.tex && pdflatex main.tex
```

The Python stack is `numpy` + `scipy` + `matplotlib` only; no neural network or stochastic training code appears anywhere in the pipeline.

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